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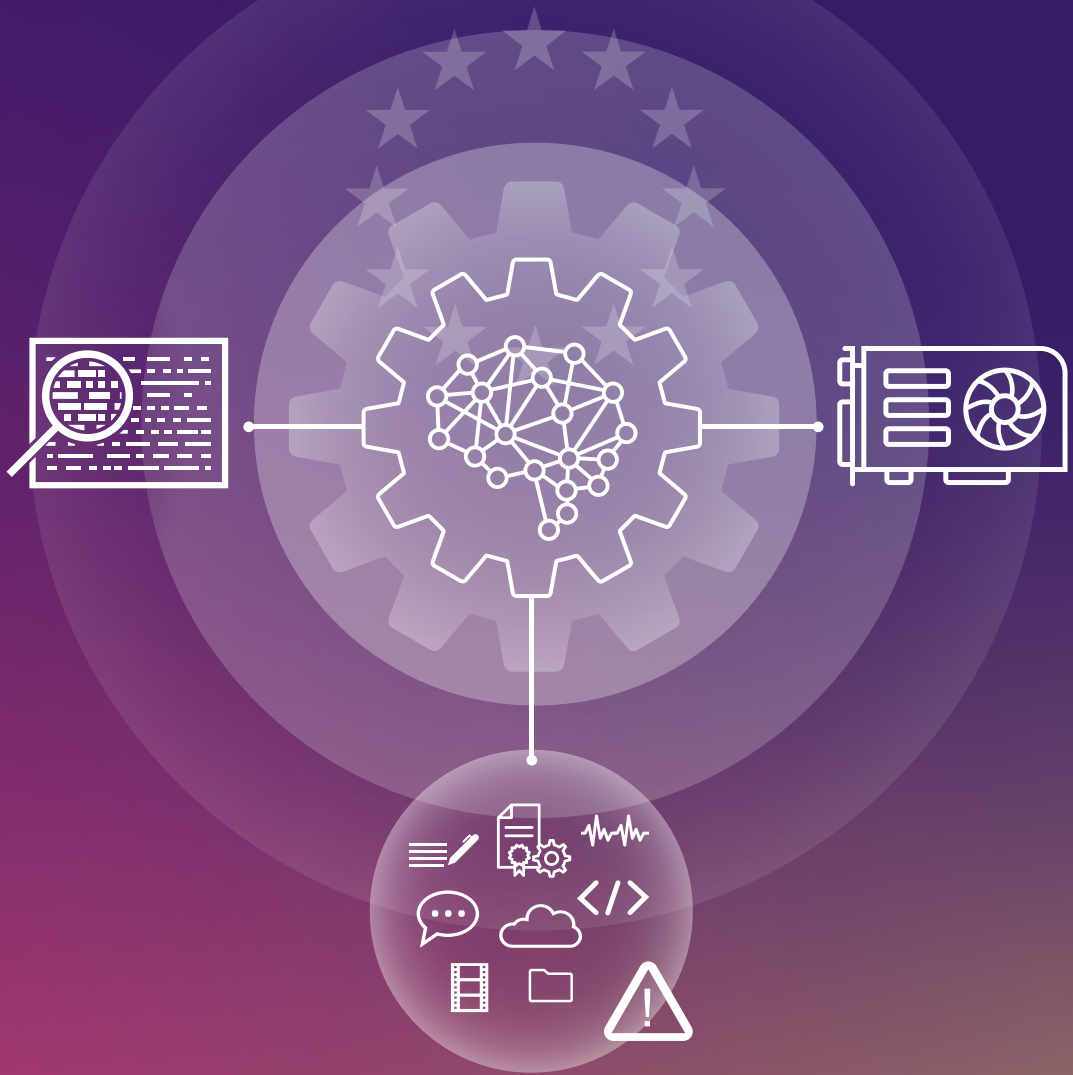
Training Compute Thresholds - Key Considerations for the EU AI Act

Collection of External Scientific Studies on General-Purpose AI Models under the EU AI Act

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2025



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JRC143255

PDF ISBN 978-92-68-31430-2 doi:10.2760/3546833 KJ-01-25-458-EN-N

Luxembourg: Publications Office of the European Union, 2025

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How to cite this report: Erben, A., Negele, M., Heim, L., Sevilla, J., *Training Compute Thresholds - Key Considerations for the EU AI Act*, Fernández Llorca, D., Gómez, E. (editors), Publications Office of the European Union, Luxembourg, 2025, https://data.europa.eu/doi/10.2760/3546833, JRC143255.

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Abstract

This report provides an in-depth analysis of the concept of cumulative compute as a proxy for general-purpose AI (GPAI) model capabilities, with a focus on measuring and verifying training compute, defined as the computational resources used to train a model, measured in floating-point operations (FLOP). It presents two approaches to estimating training compute, namely hardware-based and parameter-based methods, and discusses their strengths and limitations, including the challenges of estimating training compute for complex architectures and the need for standardised methodologies. The report also explores the challenges of verifying declared and undeclared training runs, and discusses potential verification methods, including concordance between measurement approaches, whistleblower protection, and monitoring of large compute clusters. Additionally, it examines the regulatory context, including the EU AI Act, and provides guidance on notification triggers, including the proposed notification point at the pre-training resource commitment stage, and the need for additional clarity for what constitutes reasonable certainty about threshold exceedance. The report also discusses the importance of updating training compute thresholds to maintain their effectiveness, and proposes a framework for dynamic threshold adjustment, including regular review periods and ongoing assessment by regulatory bodies and expert groups.

Acknowledgements

The authors would like to thank Guillem Bas Graells, Jonas Schuett, Michael Chen, Michal Kubiak, Pablo Villalobos, Sophie Williams and Mauricio Baker, for their helpful input and feedback.

The editors would like to thank the JRC colleagues who have helped us develop the work resulting in this collection, the AI Office for its inputs, as well as those who have kindly agreed to review the drafts and the final external study reports.

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Note from the Editors

The EU AI Act entered into force on 1 August 2024, with the aim of promoting innovation in and uptake of AI in the Union, while ensuring a high level of protection of health, safety and fundamental rights, including democracy and the rule of law. Chapter V of the AI Act outlines obligations for the providers of general-purpose AI (GPAI) models, which are AI models *"trained with a large amount of data using self-supervision at scale, that [display] significant generality and [are] capable of competently performing a wide range of distinct tasks [...] and that can be integrated into a variety of downstream systems or applications"*. Moreover, the chapter specifies additional obligations for the most advanced GPAI models, those that pose systemic risks, which are classified as such according to criteria established in Article 51 and Annex XIII. From 2 August 2025, the obligations for providers of GPAI models and GPAI models with systemic risk enter into application.

The European Commission's Joint Research Centre (JRC) has been providing scientific support throughout the legislative process of the AI Act since 2020. After the Council and Parliament reached a final agreement in December 2023, the JRC initiated an internal study, which included two external experts, focusing on the technical aspects of Chapter V that likely required further clarification. This study generated an internal report titled "General Purpose AI Models under the AI Act" (Hernández-Orallo et al., 2024), which provided preliminary insights into compute, generality, capabilities, and systemic risks. One of the clear conclusions of this preliminary study was that further scientific work was necessary.

Between September 2024 and June 2025, the JRC setup and managed, in close collaboration with the EU AI Office, a pool of 15 external experts with diverse expertise and backgrounds. This expert pool produced further technical scientific input on key aspects of Chapter V, through the development of methodologies for categorising AI models as GPAI models and for classifying GPAI models as GPAI models with systemic risk, to inform implementation of the EU AI Act. The experts also provided input on the recently published Commission guidelines on the scope of obligations for providers of GPAI models (European Commission, 2025), as part of the public multi-stakeholder consultation. The primary outcome of this expert pool is this **Collection of External Scientific Studies on General-Purpose AI Models under the EU AI Act**, which comprises a total of six external scientific study reports.

Although more documents may be added to the collection in the future, as of the writing of this editorial, the titles of the external reports included in the collection are:

- Training Compute Thresholds - Key Considerations for the EU AI Act
- A Framework for General-Purpose AI Model Categorisation
- A Framework to Categorise Modified General-Purpose AI Models as New Models Based on Behavioural Changes
- A Proposal to Identify High Impact Capabilities of General-Purpose AI Models
- The Role of AI Safety Benchmarks in Evaluating Systemic Risks in General-Purpose AI Model
- General-Purpose AI Model Reach as Criterion for Systemic Risk

The overall objective of this collection is twofold. On the one hand, it aims to contribute to broadening the understanding and discussion of the technical and scientific issues related to GPAI models and the identification of systemic risks. On the other hand, it seeks to provide a solid scientific basis for informing the implementation of Chapter V of the EU AI Act, which has recently entered into application. It is clear that we are dealing with complex issues, where a clear scientific consensus has yet to be established, and which require a certain degree of flexibility. Nevertheless, this is part of the

necessary effort to promote innovation and the uptake of AI, while ensuring protection for human health, safety, and fundamental rights in Europe.

These external scientific studies cover aspects regarding the presumption of having high impact capabilities based on cumulative amount of computation used for training (Article 51(2)), notification conditions (Article 52), the definition of a GPAI model (Article 3(63)), and considerations for GPAI models being classified as GPAI models with systemic risk based on capability benchmarks, safety benchmarks and reach (Article 51(1) and Annex XIII).

These studies reflect the outcome of the scientific and technical analysis of a series of external experts to the Commission. In some cases, they present a state-of-the-art review, while in others, they propose methodologies based on solid scientific evidence, while acknowledging significant uncertainty, as many of these problems still lack a widely accepted solution. The content, analysis, recommendations, and suggestions should then not be interpreted in any way as the position of the Commission, nor of the JRC editors in particular, but rather as the opinion of the authors.

Executive summary

This executive summary summarises our work on training compute thresholds while highlighting key questions and challenges in their implementation. Some challenges can be addressed by clarifying policy goals, while others require technical details that are currently unavailable—illustrating challenges of a "model-focused" regulatory approach. Our aim is to provide practical guidance while acknowledging uncertainties in the field. We highlight our current suggestions in blue boxes and outline key challenges that require further discussion in red boxes. We provide an exemplary guidance for measuring training compute in Annex 1.

Background and Context from the EU AI Act

The EU AI Act (European Union, 2024) establishes a regulatory framework for AI, with special attention to general-purpose AI (GPAI) models that may pose systemic risks. According to Article 51(2), a GPAI model is presumed to have "high-impact capabilities" (capabilities that match or exceed those of the most advanced models) and thus could pose systemic risk if its "cumulative amount of computation used for training measured in floating point operations is greater than 10^{25}." Providers of models that exceed this threshold must comply with various obligations, including notifying the European Commission (Article 52), performing model evaluations, assessing and mitigating systemic risks, reporting serious incidents, and ensuring cybersecurity (Article 55). Moreover, Article 51(3) provides that this threshold, as well as benchmarks and indicators for assessing high-impact capabilities, should be updated in light of technological developments.

For effective implementation, some aspects may require additional clarifications:

- Which models should be captured by these thresholds.
- When thresholds should be updated.
- When and how providers should notify authorities.
- How to verify compliance with reporting requirements.
- How to measure training compute.

These questions are difficult to answer from a purely scientific perspective[1]. They also require understanding the goals of the regulation and practical implementation constraints. Given uncertainties, such as the upcoming Codes of Practice, secondary legislation and evolving training methodologies, we cannot provide conclusive answers to all questions. Instead, this paper provides technical guidance to inform decision-making based on policy goals and techno-scientific realities.

What Are Training Compute Thresholds

Training compute thresholds use the total computational operations required to train an AI model as a proxy for its capabilities and potential risks. Training compute offers several key advantages for regulatory purposes:

1. **Capability-tracking**: Empirical research has established correlations between training compute and model capabilities through scaling laws. As models become more capable, they may pose greater risks from misuse or pursuing misaligned objectives.

[1] Additionally, developing guidance on compute thresholds is challenging due to limited information of AI companies' internal processes and the continuous developments in the field. If we had written this paper a year ago, our recommendations would likely have been different. The rapid evolution of the field illustrates this point perfectly—we have recently observed reasoning-specialised models from OpenAI, Anthropic, DeepSeek, and others, but with the exception of Chinese company DeepSeek, none have shared detailed information about their training methods or compute allocation. Consequently, much of our guidance is based on extracting general lessons from limited data and from informal discussion with employees of AI companies.

2. **Measurable before deployment**: Training compute can be estimated before development and known before deployment.
3. **Externally verifiable**: External experts can verify the measurement, and also third parties, such as compute providers, can potentially verify usage without accessing proprietary information.
4. **Cost-tracking**: The amount of compute used correlates with the financial resources required, focusing regulation on the most resourced actors.

Training compute is quantified by counting the number of operations (typically floating-point operations or FLOP) used throughout a model's training process. We discuss guidance on how to measure training compute in Annex 1.

While this relationship is not perfect—algorithmic improvements continuously enable more capabilities with less compute—it currently represents our best available proxy for measuring potential model capabilities before deployment; other metrics score worse on the required dimensions. A regular review process would help to assess its effectiveness over time.

Our suggestion for how training compute thresholds should be used:

— We recommend using training compute thresholds specifically for identifying frontier AI models/systems —those at the cutting edge of capabilities. Rather than attempting to track specific capability levels over time (which becomes increasingly difficult due to algorithmic efficiency improvements), compute thresholds are most effective when used to identify the top 10-20 most advanced models for regulatory scrutiny.

— We recommend implementing an inclusionary-by-default framework for defining "cumulative compute" that counts all computation directly contributing to parameter updates in deployed models, while allowing specific exclusions based on clear decision criteria. See Annex 1.

Challenges for training compute thresholds:

A. The correlation between pre-training compute and capabilities has weakened as AI model/system lifecycles evolve. Historically, capability gains came primarily from a distinct pre-training phase, but this relationship is becoming more complex. Modern systems blur these boundaries through techniques like “test-time” compute (as in OpenAI’s o1) that enhance reasoning during inference, synthetic data generation from other AI models, and emerging training paradigms like “self-play” (demonstrated in game AI like AlphaZero and now emerging in language models) that rely heavily on inference-style computation. Additionally, frequent model updates (particularly evident in OpenAI’s model family) make it increasingly difficult to distinguish between genuinely new models and modified versions. These developments fundamentally change how training compute translates to capabilities and challenge our ability to identify distinct lifecycle phases that produce the most significant capability gains.

B. The term "cumulative compute" in the EU AI Act creates measurement challenges that will benefit from further clarification. While we provide guidelines for attributing compute for known techniques (distillation, mixture-of-experts, model merging), future training paradigms will blur these boundaries further. Each regulatory definition established today inadvertently creates incentives for architectures that comply technically while circumventing the spirit of compute accounting, with the potential unintended consequence of transforming ‘cumulative compute’ into a perpetual regulatory challenge requiring frequent reassessment.

C. No standardised methodology exists for measuring training compute across different architectures and training paradigms, creating potential challenges in reporting.

Setting and Updating the Threshold

The setting and updating of compute thresholds may involve balancing at least three fundamental parameters: (i) *regulatory objectives* (which models and risks to capture), (ii) *practical constraints* (available resources at the regulator), and (iii) *market impact* (costs imposed on providers developing these models).

How to set the threshold initially

The current EU AI Act threshold of 10^{25} FLOP was likely established to capture models with capabilities matching or exceeding "the most advanced general-purpose AI models" at the time of drafting.

There is no immediate clear answer to where the threshold should optimally be set. The current 10^{25} FLOP threshold may be reasonable depending on regulatory goals—whether to capture only the most advanced frontier models or include a broader range of models with potential systemic impacts.

Why to update

Training compute thresholds will require periodic adjustments to maintain their objectives. Key factors driving these adjustments include:

— **Algorithmic Efficiency Improvements**: As algorithms become more efficient, equivalent capabilities can be achieved with less compute, potentially requiring downward threshold adjustments to maintain consistent capability coverage.

— **Societal Adaptation**: As society adapts to AI advances, the risk profile of models with given capabilities may change, affecting appropriate threshold levels. A once risky model might not be risky anymore because other AI models help defend against its risks or society has built resilience to its potential harms.

— **Assessment Capacity Constraints**: Regulatory bodies have finite assessment resources, necessitating thresholds that produce a manageable number of assessments (depending on what the regulator plans and, e.g., the type and intensity of evaluations required). Any downward threshold adjustment would significantly expand the scope of models.

These factors create the need for updates in different directions: the algorithmic efficiency factor suggests lowering thresholds over time, while societal adaptation and capacity constraints might suggest raising them.

How to update

We see two complementary potential approaches for threshold adjustment:

1. **Dynamic thresholds** linked directly to frontier model scale (e.g., one order of magnitude below largest known training runs) or using the aggregate training compute scaling trend of frontier models for more robustness.
2. **Regular review periods** with structured consultations (we suggest twice per year), including an ongoing assessment through expert bodies like the EU AI Office (with the option to call a review in light of new developments)

Notably, threshold-based classification does not need to be perfect, as the EU AI Act provides another pathway for classifying models as posing systemic risk based on EU AI Office assessment, even if they do not exceed the compute threshold, as established in Article 51(1)(b).

Our suggestion for setting and updating training compute thresholds:

— We recommend adjusting the threshold to 10^{26} FLOP, as the current 10^{25} threshold would capture too many models, including previous generation models/systems that likely do not pose systemic risks. This would focus regulatory attention on a manageable number of truly frontier models where evaluation capacity can be effectively deployed.

— For updating the threshold, we suggest biannual reviews conducted by the enforcement authority with input from technical experts. These reviews should consider: (1) the number of models captured by the current threshold, (2) observed capability gains in recent models, and (3) evidence of algorithmic efficiency improvements.

Challenges for setting and updating training compute thresholds:

A. Is the primary goal to identify models first crossing the threshold to learn about "what is coming down the pipeline", or should thresholds capture all models at a particular capability level (which would require updating downward as algorithms become more efficient)?

B. How will risk be measured and evaluated in contrast to raw capability? The relationship between capabilities and systemic risk remains poorly understood.

C. What is the expected capacity of the enforcement authorities, and how many models can they realistically evaluate thoroughly? Clearer guidance on the evaluation procedure and capacity will inform notification timing and threshold recommendations.

D. Given that algorithmic efficiency is critical for tracking against compute growth trends, but gathering empirical evidence through experimentation is prohibitively expensive, could the enforcement authorities establish dedicated resources for conducting targeted ablation studies?

When to First Notify

The EU AI Act requires providers to notify the Commission when their model meets the compute threshold for presumption of systemic risk, or within two weeks after it becomes known that this will be the case (Article 52(1)). This raises a critical question: at what point in development can a provider reasonably be expected to "know" that a model will exceed the threshold?

While development processes vary across organisations and remain largely proprietary, we can make reasonable inferences based on fundamental economic principles—when organisations invest significant resources in compute (and therefore money), they necessarily engage in substantial planning. Based on limited public information about model development timelines, we suggest that developers would typically **have sufficient certainty to know that a model will exceed the threshold approximately one to three weeks before the final pre-training begins**. However, the information available at this stage may have limited value for risk-assessment purposes as the model is not trained yet.

When providers notify the Commission, we recommend they include data for compute verification: training duration, cluster specifications (accelerator count and type), expected utilisation rates, and basic architectural parameters. While the enforcement authority will likely need to develop detailed reporting requirements based on their assessment needs, these metrics allow verification of the compute estimation.

Our suggestion for when to notify:

— We recommend a tiered notification approach: an initial notification when compute allocation decisions are finalised (typically 3-4 weeks before training begins), followed by a more detailed submission once training is complete, and later again once the model is finalised.

— This early notification provides regulators with advance awareness while acknowledging that comprehensive risk assessment information will only be available after training is complete.

— The specific information requirements should be aligned with the assessment objectives and capacity constraints of the enforcement authorities.

Challenges for when to notify:

A. As training paradigms evolve (particularly with continuous training and multi-stage approaches), the concept of a discrete "training run" becomes increasingly ambiguous. This complicates notification timing requirements, as the notification timing depends on the regulator's assessment objectives and whether they require a fully trained model for evaluation or seek to intervene earlier in the development process.

B. The question of "when is a modified model a new model?" becomes critical for notification requirements. Does fine-tuning or updating an existing model trigger new notification obligations? See (Berglund et al., 2024).

C. What specific information would actually be useful for the enforcement authority to receive at the notification stage, and what kind of assessment do they want to conduct at this point? Which information is available to the provider at the point of notification (e.g., before they have trained the model)?

D. Which processes must the enforcement authority complete before conducting a full model evaluation, and what advance information would facilitate preparation? The notification timing requirements should be informed by the regulator's operational needs. For example, if setting up evaluation infrastructure and allocating expert resources requires four weeks of lead time, this would suggest notifications should occur at least that far in advance of model completion. Understanding these practical constraints and preparatory requirements would help establish notification timelines that serve both regulatory and developer needs.

Validating and Verifying GPAI Model Reporting

The EU AI Act requires providers to notify the Commission when their models cross the compute threshold and to document their training compute usage. We find two theoretical verification challenges:

1. **Verifying declared amounts of training compute**: Ensuring providers accurately report their training compute.
2. **Detecting undeclared training runs**: Identifying providers who fail to report large training runs.

To verify declared amounts of training compute, we find the following verification options:

1. **Concordance between measurement approaches**: Comparing hardware-based and parameter-based estimates (Annex 1) can help identify inconsistencies. Significant divergence

between these methods, or misalignment with other disclosed information, may indicate non-compliance.

2. **Compute usage verification through cloud service providers (CSPs)**: CSPs have access to metadata on computational workloads that could estimate training run size. Obtaining this information may require new legal frameworks or voluntary cooperation.

To find undeclared training runs, we find the following verification options:

1. **Know Your Customer (KYC) requirements for cloud providers**: Cloud providers could be required to identify entities with access to compute above specified thresholds, similar to anti-money laundering frameworks in the financial sector.
2. **Monitoring of large compute clusters**: Regulators could create and maintain a database of known compute clusters, tracking their size and ownership. Companies known to possess large clusters would be candidates for training models above the threshold.
3. **Performance-based verification**: Inference of training compute from benchmark performance could help identify models that likely exceeded the threshold without proper notification, leveraging established scaling laws.
4. **Whistleblower protection**: Enabling employees of GPAI providers and Cloud Service Provides (CSPs) to safely report compliance violations could provide an additional verification mechanism.

Given the global nature of AI development and compute infrastructure, most of these approaches would require international coordination. However, in practice, the feasibility of comprehensive verification is likely limited without additional legal frameworks or international agreements.

Our suggestion for validating and verifying training compute usage:

— Require dual reporting using both hardware-based and parameter-based estimation methods.

— Leverage the compute-to-performance relationship to identify outliers—models performing unusually well for their reported compute would trigger further investigation.

— Establish protected whistleblower channels.

Challenges for validating and verifying training compute usage:

A. Legal authority for accessing cloud provider data or implementing KYC requirements for compute usage remains unclear under the current EU AI Act framework.

B. There is an inherent tension between verification stringency and market impacts: highly intrusive verification mechanisms could be perceived as heavy-handed.

C. CSPs may resist implementing verification mechanisms that could compromise client confidentiality or proprietary information.

Challenges and Limitations

Throughout this report, we have identified specific challenges within each area of compute threshold implementation. Several recurring themes emerge that warrant particular attention:

— **The Blurring Definition of "Model"**: As training methodologies advance, the concept of a discrete "model" becomes increasingly ambiguous. When does fine-tuning create a new model? How should we classify models that incorporate components from other models? These questions complicate both measurement and notification requirements.

— **Training vs. Deployment Convergence**: The boundary between training and deployment is rapidly dissolving. Techniques like test-time compute (used in OpenAI's o1), synthetic data generation, and self-play methods increasingly blend what were once distinct phases.

— **Capability vs. Risk Relationship**: While compute correlates with capabilities, the relationship between capabilities and actual risks is dynamic, not static. A capability that initially poses significant risks may become less concerning as society develops resilience mechanisms, countermeasures, or defensive AI applications.

— **Future-proof Cumulative Compute Accounting**: Emerging training paradigms (model averaging, distillation, training-time-inference, multi-model interactions) require a stable definition of "cumulative compute" for regulatory and market confidence. While we provide a foundational framework, creating numerous specific rules for each technique is impractical. Novel approaches may rapidly emerge within less than a year, potentially undermining regulatory objectives (identifying top-tier models, proportionate compliance requirements, maintaining FLOP count comparability).

1 Introduction

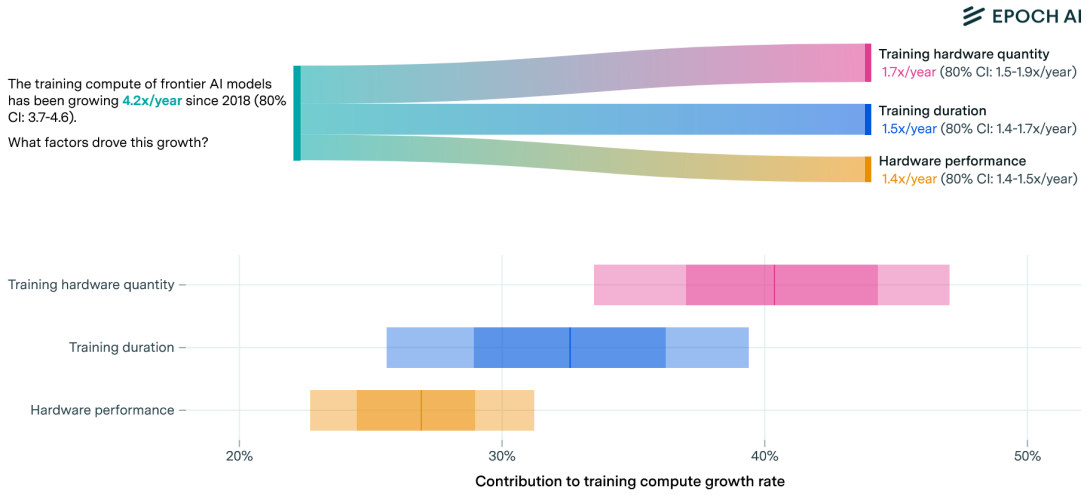
Preventive governance of general-purpose AI (GPAI) models presents a fundamental regulatory challenge: implementing oversight mechanisms before deployment while preserving innovation. This challenge is particularly relevant given AI risks can be severe or irreversible once deployed (Hendrycks et al., 2023).

Effective governance requires technical indicators that can reliably anticipate significant capability advances. These indicators must be both measurable before deployment and resistant to gaming, while maintaining compatibility with rapid technological progress.

Training compute—the computational resources invested in developing an AI model, measured in floating-point operations (FLOP)—is one technical indicator that can be used for preventive governance. Three properties make training compute particularly suitable:

1. Pre-deployment quantification: Training compute requirements can be estimated during the planning phase, before training.
2. External verification: Hardware usage can be verified without accessing proprietary model details.
3. Circumvention resistance: Reduced training compute fundamentally constrains model capabilities, therefore, under the current training paradigm, training compute is indicative of model capabilities and this relationship is not easily broken.

Figure 1: Decomposition of training compute growth since 2018, split into three similarly influential components. The box plot displays 80% (darker) and 50% (lighter) confidence intervals.



Source: EpochAI (Emberson and Owen, 2025b).

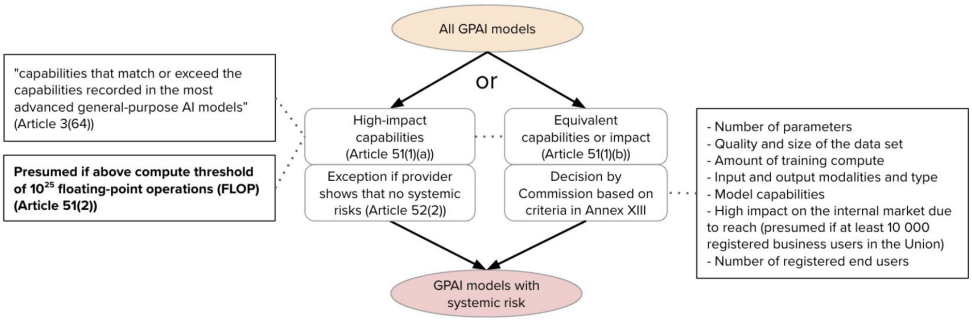
Training compute for notable AI models has grown dramatically, with a four- to five-fold annual increase between 2018 and 2024 (Figure 1). While this growth traditionally stemmed from advances in hardware performance, extended training durations, and increased hardware quantity, recent trends show a shift in their relative importance. Hardware quantity dominates this growth trajectory (Emberson and Owen, 2025a) while other factors continue their upward trend—as seen in developments in the price-performance of leading AI chips—but at a slower pace (Rahman, 2024).

Training compute functions as a leading indicator for frontier capabilities. Neural scaling laws (Wikipedia, 2025) demonstrate that increased compute consistently leads to better model performance and likely novel capabilities. Even when algorithms become more efficient, the industry

trend has been to reinvest those efficiency gains into larger models rather than reducing overall compute spending (Barnett, 2025). If leveraged correctly, this relationship could enable regulators to react to capability advances.

The properties of training compute as a governance metric have translated into concrete regulatory implementation through training compute thresholds in the EU AI Act. According to Article 51(1) a GPAI model is classified as posing systemic risk if it has "high impact capabilities" (i.e., capabilities that match or exceed those of the most advanced models, Article 3(64)) or "capabilities or an impact equivalent to those set out in Article 51(1)(a)". If the cumulative training compute for a model, measured in FLOP, exceeds the threshold of 10^{25} FLOP, then according to Article 51(2) a model "shall be presumed to have high impact capabilities pursuant to paragraph 1, point (a).". Figure 2 summarises the classification methodology.

Figure 2: Systemic risk classification of GPAI models in the EU AI Act.



Source: (Heim and Koessler, 2024).

However, implementing this governance mechanism creates several technical challenges. The first concerns precise measurement and definition: while the EU AI Act provides initial guidance in Recital 111—defining cumulative training compute as computational resources used in "activities and methods intended to enhance model capabilities prior to deployment"—determining which training activities contribute to capability enhancement remains complex.

The relationship between training compute and capabilities presents a second challenge. Developers find innovative ways to train more efficiently, whether through higher-quality data, more efficient architectures, or novel training methodologies. This means that for a given capability level, the amount of required training compute steadily decreases[2]. Such advancements challenge the ability to maintain training compute as an effective capability proxy because a fixed training compute threshold will capture increasingly capable models over time. What once required extensive computational resources may later be achieved with significantly less training compute while delivering equivalent or superior capabilities.

This changing correlation directly affects how capabilities relate to specific training compute levels over time. If regulators aim to target models with consistent capability levels, thresholds would need corresponding adjustments. However, the direction of these adjustments is not straightforward, for example, as society adapts to AI advances, risk profiles may shift, potentially altering the appropriate regulatory scope. Establishing a transparent method for threshold adaptation would be valuable, allowing to maintain effectiveness as the relationship between compute and capabilities evolves.

Finally, practical measurement and verification of compute expenditures introduces additional complexity. Hardware utilisation metrics and model architecture characteristics offer potential pathways, but each approach involves trade-offs between accuracy, verifiability, and protection of proprietary information.

[2]This trend is likely to hold true regardless of model size as more efficient techniques can be applied at all model size scales. However, it is worth noting that model size may stop increasing due to economic viability, which applies a downward pressure on compute spendings but is currently not strong enough to revert our trend (Narayanan and Kapoor, 2024)

This study develops two complementary technical frameworks for compute-based governance along with practical implementation guidelines. First, we propose a framework for defining cumulative compute that maintains its validity with technological progress as a capability proxy (Section 2), and evaluate the different methodologies for estimating training compute (Section 3). Our second framework addresses the question of how to define and dynamically update cumulative compute thresholds (Section 4). Building on these foundations, we analyse notification timing considerations for providers anticipating their GPAI models will reach specific FLOP thresholds (Section 5) and explore verification mechanisms for training compute declarations (Section 6). We conclude by examining limitations and open research questions regarding training compute effectiveness as a regulatory instrument (Section 7).

2 Cumulative Compute as a Capability Proxy

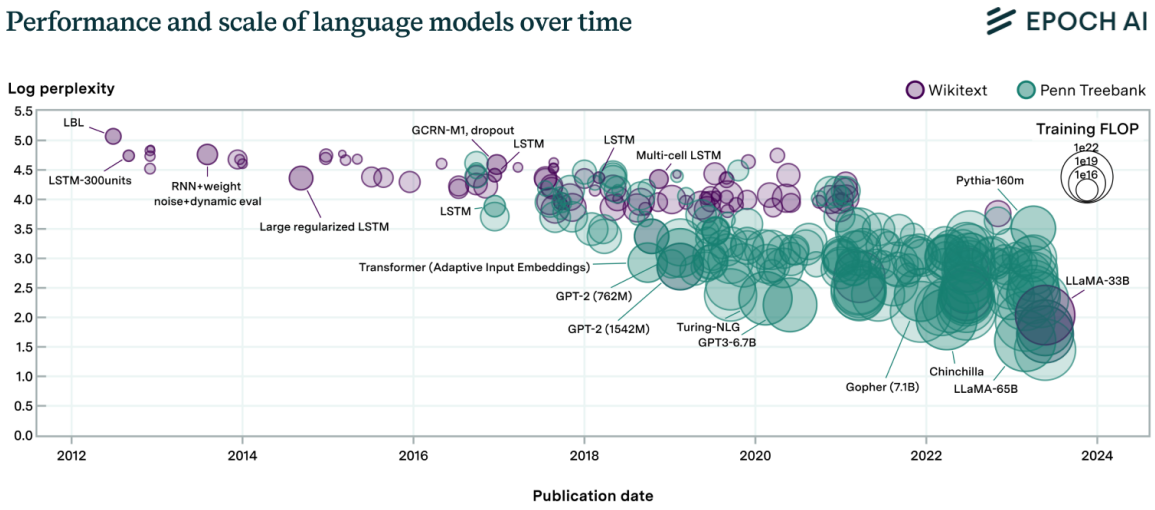
The implementation of compute-based thresholds for AI governance requires a concrete definition of "cumulative compute". While Recital 111 provides initial guidance through its focus on capability-enhancing computational activities and methods, the scientific community lacks consensus on which computational activities should be included.

Given this context, we aim to develop a definition tailored to the needs of the EU AI Act: allowing for the presumption of high-impact capabilities and potentially systemic risk before deployment. To address this challenge, we must examine the relationship between compute and capabilities to inform an effective definition.

2.1 Background: Compute as a Predictor for AI Capabilities

At the frontier of AI development, training compute serves as a valuable governance metric due to its demonstrated relationship with model capabilities (Heim and Koessler, 2024). Pre-training compute shows a strong correlation with model performance in predicting the next token (cf. Figure 3), as measured by model loss (log-perplexity), which is systematically minimised during pre-training (Ho et al., 2024). This statistical relationship forms the foundation of established scaling laws that enable capability forecasting based on compute allocation.

Figure 3: Performance of 231 language models (measured in log-perplexity, lower is better) against their publication date and scale (measured in FLOP), showing a clear trend with larger runs achieving a lower loss.



Source: EpochAI (Ho et al., 2024).

Post-training Compute and Specialised Capabilities. Beyond pre-training, the correlation between compute and capabilities becomes more complex. Post-training techniques such as reinforcement learning from human feedback (RLHF) or supervised fine-tuning (SFT) can substantially enhance specific capabilities with comparatively modest compute investments. Current evidence suggests that compute weakly correlates with improved performance aggregated over many individual tasks, allowing the prediction of performance in existing benchmarks (Owen, 2024, Pimpale et al., 2025). However, compute is not a reliable predictor for individual task performance, suggesting that the choice and implementation of specific techniques may be more determinative for specialised capabilities.

Compute Allocation Between Pre-training and Post-training. Current industry practice reflects

the importance of pre-training, with the majority of training compute typically allocated to pre-training[3]. This allocation pattern suggests that extensive pre-training remains a prerequisite for successfully applying post-training techniques in state-of-the-art models. However, this paradigm is evolving. DeepSeekR1's allocation of approximately 20% of its compute budget toward post-training to enhance reasoning capabilities signals a potential shift in compute distribution (Erdil, 2025). Had this analysis been conducted a year ago, we likely would not have anticipated this significant reallocation toward post-training compute, highlighting the rapid evolution of training paradigms.

Critical Thresholds for Capability Emergence. The minimum amount of pre-training required for post-training techniques to work effectively remains an active research topic. Previously, models with approximately 34 billion parameters were required to reach strong performance (80% accuracy) on GSM8K, a standard mathematical reasoning benchmark. Studies now demonstrate that specialised fine-tuning with high-quality synthetic data can elicit similar performance levels in significantly smaller models. For instance, a 1.3 billion parameter model improved from a baseline accuracy of 45% to 68% with minimal additional compute (approx. 2% of the original training compute), and even 125 million parameter models achieved 63% accuracy through similar techniques (Liu et al., 2023).

While these results are promising for specific benchmarks, smaller models appear to lack the cross-domain reasoning flexibility exhibited by larger models. Very recent research suggests that approximately 1.6 billion parameters may represent a current "critical model size" threshold where reasoning capabilities significantly improve across a broader range of benchmarks (Hsiao and Dutta, 2025). This evolving understanding of parameter and compute thresholds complicates governance approaches that rely on fixed metrics.

Selective Pressure on Compute-efficient Techniques. Regulatory thresholds inherently create selective pressure toward compute-efficient approaches, potentially steering the field's technical development. This pressure may inadvertently discourage research into promising techniques that are currently compute-intensive but could yield long-term benefits. Model merging illustrates this challenge: when two independently trained models are combined, their compute cost could accumulate while their capabilities will not increase proportionally (Khalifa et al., 2024). Regulators must consider how frontier techniques are already employed in industry practice, for instance, merging parallel training branches is standard at some providers (Llama Team, AI at Meta, 2024). We recommend that regulatory frameworks include specific guidance on handling such cases and avoiding unintended consequences on valuable technical approaches.

Implications for Defining Cumulative Compute. Two perspectives exist regarding compute's relationship to novel capabilities. First, historical evidence suggests compute correlates with capability advances as the eight order-of-magnitude increase in compute over the past decade corresponds with significant advances in language model functionality. Second, not all novel capabilities are tightly correlated with compute investments; certain capabilities, particularly reasoning, have emerged without proportional compute scaling[4]. While this relationship seems to vary across capabilities, sufficient pre-training compute appears to establish a necessary foundation upon which capabilities can emerge when appropriate post-training techniques are applied. This relationship,

[3]Most model developers do not publish detailed training compute analyses on different training phases, but we can make an educated guess on SFT expenses with the Llama 3 405B model as an example for a recent frontier-level model. Assuming an average input and output token count of 1,000 (Table 7, (Llama Team, AI at Meta, 2024)) and using the parameter-based 6ND rule (cf. Section 3.2), a 405B-parameter model requires roughly 2.43e15 FLOP per sample during training. With about 9,000 training steps (Section 4.1.3, (Llama Team, AI at Meta, 2024)), an educated guess of the batch size being 16,000 from 16M tokens/batch during pre-training (Table 4, (Llama Team, AI at Meta, 2024)) equals 3.5e23 FLOP, only about 1% of the pre-training compute cost of $\sim$3.8e25 FLOP.

[4]"Without proportional compute scaling" refers to two observations: 1) CoT reasoning can be elicited from relatively small models (e.g., Hsiao and Dutta's 1.6B parameter model with task-specific fine-tuning (Hsiao and Dutta, 2025)), showing reasoning does not necessarily require a lot of absolute compute; 2) For structured reasoning with <think> tags, we have seen that a 20% additional compute investment in fine-tuning yields disproportionately large improvements relative to the 80% invested in pre-training—though this single reference point may not be representative of all reasoning capabilities (DeepSeekR1). This suggests a non-linear relationship where targeted compute investments can unlock reasoning capabilities efficiently once a sufficient foundation exists.

though imperfect, currently represents our best available proxy for capability development at scale.

This measurement complexity creates a fundamental tension between regulatory objectives and implementation realities. From an industry perspective, developers require regulatory stability to support strategic planning, particularly for initiatives involving potentially multi-year roadmaps. While model training typically completes within months (Sevilla et al., 2022a), the subsequent deployment, integration, and commercial application of these systems extends over longer time-frames. A clear and predictable definition of cumulative compute reduces the risk of sudden restrictions while accounting for emerging training techniques[5].

From a regulator's perspective, it is useful that compute thresholds function effectively regardless of how training processes may evolve. As the field advances, new techniques emerge that may shift computational patterns without necessarily changing the relationship between total compute investment and resulting capabilities. A framework based primarily on specific techniques would require constant revision to maintain effectiveness, potentially lagging behind technological developments.

The rapid evolution of training paradigms introduces inherent uncertainty in predicting future developments. When designing compute-based governance frameworks, we must account for these potential shifts to ensure the framework remains robust and relevant. The governance mechanism should accommodate changing relationships between compute allocation across different training phases. At the same time, it must maintain focus on regulating systems that approach capability thresholds of concern.

Building on this background, we now turn to developing a concrete classification framework for defining cumulative compute that balances scientific understanding with regulatory practicality. This framework aims to achieve three objectives:

1. Provide regulators and industry with a clear way to include or exclude training techniques aligned with capability assessment goals.
2. Limit excessive tracking burdens on developers by focusing on significant computational investment.
3. Offer a flexible toolbox of guiding principles that regulators can use to maintain governance objectives as technology evolves.

2.2 Compute Classification Framework

Effective compute-based governance requires balancing two priorities: providing model developers with clear, consistent guidelines while ensuring the threshold remains a reliable indicator of frontier capabilities as training methodologies evolve. This presents a fundamental challenge: no regulation can feasibly predict future training innovations to update cumulative compute definitions preemptively, while shifting this predictive burden to developers creates unnecessary regulatory friction. Given the compute threshold's focus on presuming high-impact capabilities, we recommend an inclusionary by-default approach to training compute classification—where all potential components initially count toward regulatory thresholds unless specifically exempted—based on the following considerations:

— **Technological Evolution**. AI training methodologies are rapidly advancing. Current patterns where pre-training dominates training compute usage may shift dramatically as new

[5]For example, a provider might employ a proprietary training technique that falls outside the current definition of cumulative compute, planning deployment without addressing systemic risk requirements. Should the definition change between training initiation and model deployment, this strategy becomes untenable, necessitating reconsideration of deployment plans and reassessment of whether the model still serves its intended purpose under the revised regulatory framework.

techniques emerge. The definition of cumulative compute needs to be amenable to these changes without requiring continuous revisions.

— **Measurement Consistency**. Both industry stakeholders and regulatory bodies benefit from a system that produces consistent, comparable measurements across different organisations and development approaches. The framework must align this goal with the reality of using simplifying assumptions when performing hardware- or parameter-based estimates (cf. Section 3), especially given the non-deterministic nature of training on large-scale systems.

Based on these considerations, our proposed framework to decide on whether a technique warrants inclusion towards the definition of cumulative compute anchors around three rules:

— **(R1) Default Inclusion with Clear Exclusion Criteria**. All computation directly contributing to parameter updates in the final deployed model is presumed to contribute to its capabilities and therefore included in its cumulative compute. This creates a clear starting point for model developers when assessing whether to monitor techniques. Conversely, operations may be excluded from measurement only when they do not directly contribute to parameter updates of the deployed model.

— **(R2) Dual Estimation Methodology**. Model developers must report compute costs using both hardware-based and parameter-based estimation methods. The final determination for regulatory threshold compliance will use the geometric mean of these two estimates, providing a more robust measurement that accounts for methodological differences.

— **(R3) Practical Significance Threshold**. To maintain regulatory focus on substantive computational investments while reducing administrative burden, operations consuming less than 20% of total training compute cumulatively may be excluded from detailed tracking requirements. However, any techniques should still be explicitly counted toward the cumulative compute threshold if they are specifically requested for detailed analysis by the enforcement authority based on risk assessment concerns.

To apply R1 in practice, we have developed decision criteria (see Table 1) that help determine whether a particular training technique directly contributes to parameter updates and should be included in cumulative compute estimates.

For any ambiguous technique, determine whether to include the computation by answering these questions:

1. Does the computation directly update the final model's parameters?
2. Does the computation create model-specific inputs that depend on the current model state?

If the answer to ANY of questions 1 or 2 is "yes", the computation would be included in cumulative compute calculations within our proposed framework. Otherwise, it would be excluded. This approach focuses on operations that meaningfully contribute to model capability development while excluding those that could be reused across different training pipelines—similar to how we recommend not counting computational costs for generating standard datasets, as these are transferable resources rather than model-specific computational investments.

Our dual estimation methodology (R2) establishes a more robust signal by requiring model developers to report compute costs using both hardware-based and parameter-based estimation methods. Under this framework, the final determination for regulatory threshold compliance would use the geometric mean of these two estimates, providing a more conservative measurement that accounts for methodological differences.

Table 1: Decision criteria for techniques on whether they directly contribute to parameter updates. Inclusion criteria, rationale, and examples to apply rule R1 in practice.

Inclusion Criteria	Rationale	Examples
Direct Parameter Updates: Operations directly updating the final model's weights.	These operations form the core of model training and are unambiguously responsible for enhancing model capabilities.	- Standard gradient descent and adaptive learning methods (SGD, Adam, etc.) - Partially-frozen model updates. - Gradient accumulation.
Model-Dependent Processes: Computation creating inputs for gradient updates that depend on the current model state or processes intrinsically linked to the training loop.	These operations generate training signals (data, labels, logits, feature maps) specifically tailored to the model's current capabilities or are inseparable from the capability development process. They can not be trivially reused for other training pipelines.	- Self-play with RL (similar to AlphaGo). - Adversarial example generation (GANs). - Knowledge distillation with a teacher that provides soft labels. - Active learning (dynamic data selection based on model's current performance).

Source: Own elaboration.

Our analysis of model compute estimates revealed an average difference of approximately 1.4x between methodologies with a standard deviation of 0.68 (cf. Figure 4). Using the geometric mean reduces this difference to approximately 1.18x (approx. 20%), balancing both methodologies without favouring either[6]. The geometric mean properly handles FLOP count ratios while maintaining symmetry regardless of which estimate is higher.

This dual-reporting approach enables cross-validation between methods. Significant divergence between parameter-based and hardware-based estimates could signal potential undercounting or methodological errors that warrant closer examination. While requiring both estimation methods does increase reporting requirements for providers compared to single-method approaches, this additional burden could be justified by the substantially enhanced transparency it provides to regulators. Hardware-based estimates typically capture all computation regardless of classification, serving as a natural ceiling check, while parameter-based estimates offer consistency across architectures, providing a more complete picture than either method alone.

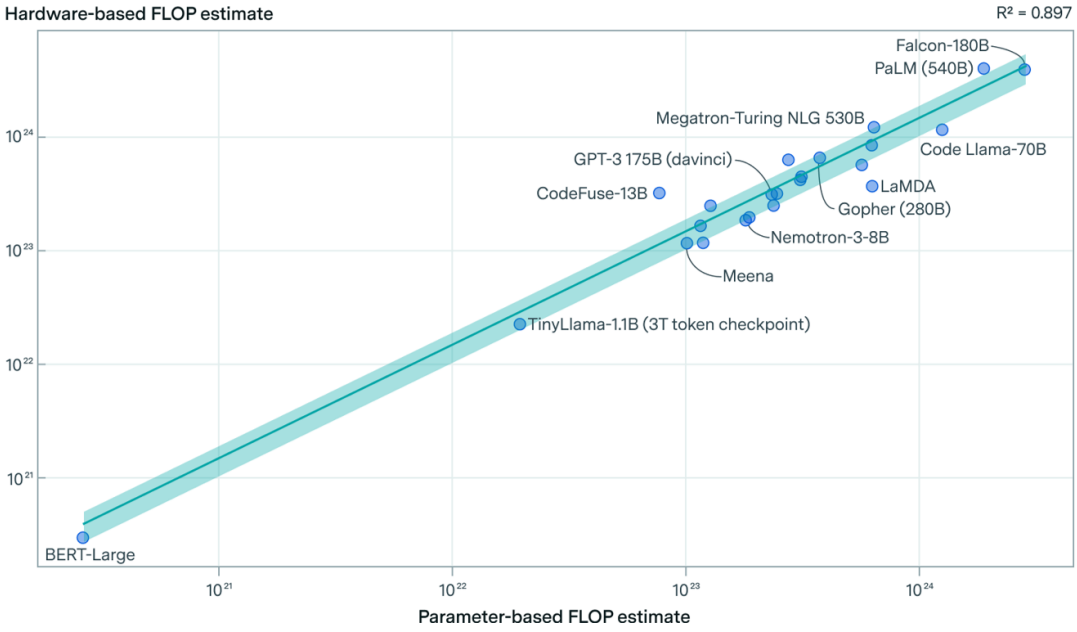
To balance comprehensive oversight with practical implementation, we have established a practical significance threshold (R3) that recognises that not all computational operations warrant detailed regulatory scrutiny. Operations consuming less than 20% of total training compute cumulatively may be excluded from detailed tracking requirements, maintaining focus on substantive computational investments while reducing administrative burden.

We established this 20% threshold by rounding up from the error margins we observed between compute estimates after applying the geometric mean, setting it conservatively above the measurement noise floor. Techniques consuming less than this threshold fall within this margin and we believe could reasonably be excluded without compromising regulatory objectives. For perspective, even a 200% overestimate would only effectively change a 2e25 training run into a 4e25 run, while compute thresholds currently operate on order-of-magnitude-scales (1000% or 10x differences).

For practical implementation, we recommend a threshold declaration approach where providers can

[6]For our two estimation methodologies, the geometric mean is calculated as $G = \sqrt{a \times 1.4a} = a \times \sqrt{1.4} \approx 1.183a$, where a represents the lower estimate and $1.4a$ the higher estimate.

Figure 4: Correlation between parameter- and hardware-based FLOP estimates. Regression between 23 arithmetic- and hardware-based FLOP estimates showing a 1.4x (± 0.68) difference and an R^2 fit of 0.897. Source: EpochAI.



Source: EpochAI.

certify that all non-explicitly reported techniques cumulatively represent less than 20% of total compute without requiring detailed submission of each minor component's compute cost. However, any techniques should still be explicitly counted toward the cumulative compute threshold if specifically requested for detailed analysis by the enforcement authority based on risk assessment concerns.

Regulators could verify compliance through benchmark analysis—comparing capability improvements against declared compute to identify discrepancies where performance exceeds expected results from declared resources. While detailed logs are not required initially, we recommend that providers maintain comprehensive documentation for potential regulatory review. This approach is strengthened by using both parameter-based and hardware-based estimates (via geometric mean), as hardware-based estimates typically capture all computation and provide a natural cross-check. Significant estimation divergence could signal potential undercounting and trigger closer examination.

We emphasise that any governance framework would benefit from striking a balance between comprehensive oversight and practical implementation costs. The bar for including techniques in cumulative compute definitions should be high, focusing primarily on significant compute costs while minimising regulatory monitoring burdens. Verification requirements invariably increase monitoring system complexity, potentially introducing substantial costs and negative market implications[7].

The following section demonstrates the practical application of our classification framework through illustrative case studies. We analyse AlphaGo Zero's training techniques as an example of computational inclusion under our framework, despite falling outside the "general-purpose" classification. We then examine auxiliary training models as a contrasting case where training compute attribution becomes partially intractable. This analysis is complemented by general implementation guidance for common training methodologies, providing concrete direction for training compute accounting across typical model development workflows.

[7]While current internal compute estimation practices within companies may operate with less formal documentation and traceability than would be required in a regulatory context, following regulatory requirements will likely increase complexity and necessitate more rigorous accounting procedures.

2.3 Case Study: AlphaGo Zero

AlphaGo Zero represented a significant moment in AI by achieving superhuman performance in the game of Go through purely self-play learning, surpassing all previous Go-playing systems including those trained on human expert games (Silver et al., 2017).

During training, the system used Monte Carlo Tree Search (MCTS), a heuristic search algorithm to generate high-quality board positions through extensive self-play. This process consumed approximately 5.4e13 FLOP per move, while the model training on this generated move was 2 orders of magnitude (OOM) cheaper at 1e11 FLOP.

AlphaGo Zero's training operates through a computational feedback loop where its neural network predicts moves for a board position, and MCTS then explores potential future positions to produce refined move suggestions that become training examples. This self-reinforcing cycle uses each board state and predicted moves to calculate gradients that directly update the policy network's parameters[8]. Since these operations directly contribute to parameter updates in the final deployed model, they satisfy our primary inclusion criterion and should be counted toward cumulative compute.

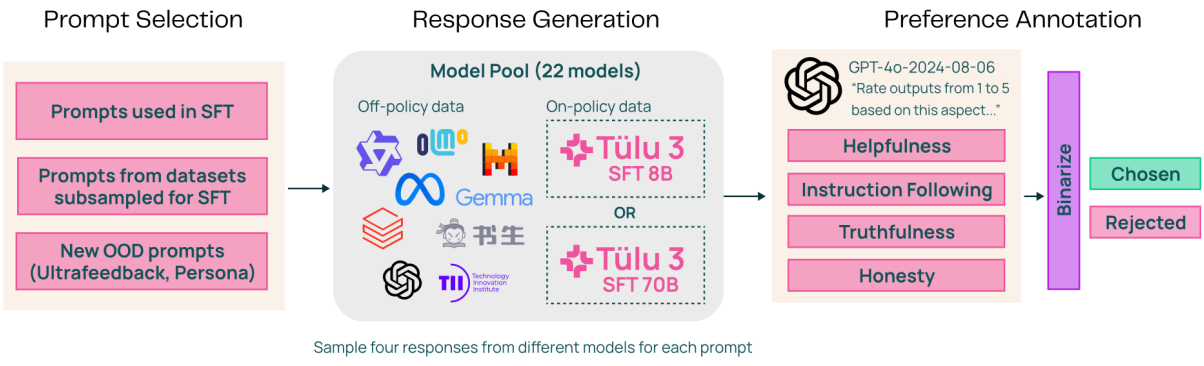
When we examine our exclusion criteria, we must ask whether the artifacts (board positions) created during training have value independent of the training process. The generated board positions derived their training value from the specific state of AlphaGo's policy at that moment. These positions would lose significant value if used to train an entirely different model due to their dependence on AlphaGo's specific policy state. This reinforces that these operations do not meet our exclusion criteria.

In summary, our framework would unambiguously classify AlphaGo Zero's self-play technique as part of cumulative compute. We now turn to the more challenging case of auxiliary models that exist in a grey area of being "directly involved" in primary model training.

2.4 Case Study: Auxiliary Models

Modern AI development often employs auxiliary models, that is, specialised AI models/systems that guide the training of other models. These auxiliary models require their own training compute and create additional inference compute costs, which raises the question: should their compute costs be counted at all, and if yes which, when measuring the total training compute of the final model? Let's examine this through the lens of preference models—auxiliary models that help make AI models better at following human preferences in conversations.

Figure 5: Pipeline for generating and scaling preference data that is based on Ultrafeedback.



Source: (Lambert et al., 2024).

[8] *Policy* in this context means learning a function that translates a state to an action, essentially trying to pick the action that maximises a certain goal (e.g., a score or a win condition) based on the current state.

First, we need to explain the training recipe of how Direct Preference Optimisation (DPO) works by looking at AllenAI's Tülu 405B model, as they have publicly shared the relevant information on how they applied DPO (Lambert et al., 2024). We start the DPO process after pre-training and supervised fine-tuning, resulting in a relatively capable model (called reference model):

1. Create a set of prompts to evaluate the model's answers.
2. Run these prompts...
   (a) ...through a pool of external models and generate answers (called "off-policy").
   (b) ...through the reference model and generate answers (called "on-policy").
3. All of these responses are graded by another capable model, e.g., GPT-4o, to get a score signalling their usefulness.
4. Use this scoring to generate pairs of "useful/not-useful" responses (note that only a subset of all responses are used in the next steps).
5. By using the reference model, these responses are translated into log-probabilities by forcing the model to output them. We save these to storage, but can also create them during training (next step) if hardware capacity permits.
6. During training, we compute the trained models' log-probabilities on the response pairs, load the reference model's log probabilities from storage, compute the DPO loss and apply the gradients to the trained model.

This process creates/requires the following artifacts:

— (A1) Set of prompts.

— (A2) Pool of external models.

— (A3) The reference model.

— (A4) The on-/off-policy responses to the prompts.

— (A5) A rating of these responses by an external model on their usefulness.

— (A6) A filtered set of response pairs by grouping "useful/not-useful".

— (A7) The log probabilities of the reference model to these response pairs.

— (A8) The log probabilities of the trained model to these response pairs.

We can roughly group them in the following categories: Reusable synthetic data (A1 + A4 + A5 + A6), AI models (A2 + A3) and non-reusable synthetic data (A7 + A8). Let's analyse them under our inclusion criteria.

(R1) Default Inclusion with Clear Exclusion Criteria. Only step 6 in the complete training setup directly affects model training through forward and backward propagation. Every other step is decoupled from the model training itself and produces artifacts (A1-A6) that can be largely re-used for other model training setups except for A7, which uses the specific reference model being trained. A8 is directly created through forward passes and used for backward passes during training, so its compute must be counted. The key question is whether to account for compute spent creating A7.

While the creation of A7 does not directly contribute to parameter updates of the deployed model (as it can be decoupled from the training process), this artifact does not retain value independent of the

training process. A7's value is tied specifically to the reference model, making it unsuitable for reuse in training other models. Therefore, following our framework, computation to create A7 should be accounted for in the cumulative compute definition—model developers need to monitor the FLOP spent during the forward passes of the reference model.

(R3) Practical Significance Threshold. The FLOP spent to compute A7 and A8 are currently negligible under typical training recipes. For Tülu 405B, the total computation consists of training FLOP (6ND) and reference model inference FLOP (2ND), totalling approximately $8 \times 150e6$ tokens $\times$ 405e9 parameters $\approx$ 5e20 FLOP[9]. This is roughly five orders of magnitude smaller than the original model's training compute of 4e25 FLOP, and the capability gains from DPO show only marginal improvement in aggregate performance evaluation (pure SFT: 58.2 vs. off+on-policy: 61.2). Under current implementations, this technique would fall below our 20% practical significance threshold, suggesting it could be exempted from detailed tracking requirements based on R3.

However, this case illustrates an important limitation of compute as a capability proxy. Despite using comparatively minimal computational resources, DPO and similar post-training techniques can significantly enhance specific model capabilities that may be particularly relevant to systemic risk considerations. If the enforcement authorities determine that these capability changes warrant the definition of "high-impact" capabilities under their risk assessment framework, they could invoke the provision in R3 to require accounting of this compute despite its small relative size. This would enable the collection of detailed technical information about emerging techniques that might otherwise escape regulatory visibility due to their computational efficiency.

We want to highlight two counterfactuals for this analysis that provide insights into the interaction dynamics between model developers and regulatory frameworks, particularly regarding computational attribution:

— First, consider a scenario with complete transparency across the training pipelines of auxiliary models and synthetic data generation. Under this scenario, we could incorporate the training compute of any auxiliary model used during the primary model's development. However, this approach faces significant practical challenges in capability attribution: it remains exceptionally difficult to quantify how the training compute invested in auxiliary models translates to quality improvements in the data they produce for the primary model.

— Second, even if capability attribution could be empirically verified—for example, demonstrating a measurable improvement in aggregate performance metrics for a specific model from the model pool—fundamental challenges remain in translating these capability improvements into proportional computational costs. This represents a broader methodological challenge in compute governance: the inability to establish reliable mapping between capability improvements and computational expenditure when capabilities derive from external sources.

2.5 Additional Considerations

After examining two case studies where we applied our framework, we want to address common considerations around training compute estimation. Our analysis aligns with many best practices established by the Frontier Model Forum (Frontier Model Forum, 2024)[10], an industry consortium focused on AI safety standards, demonstrating growing consensus on measurement approaches across the field.

Counted operations. Matrix multiplications are the primary operation used in AI training. While other operations like element-wise addition and normalisation are used during current frontier model

[9] The parameter estimation formulas are explained in more detail in Section 3.

[10] While we generally align with the Frontier Model Forum's technical measurement guidelines, we differ on certain implementation aspects outside our core topic of training compute estimation and attribution, such as fixed reporting timelines, which are beyond the scope of this analysis.

training, they make up a very small fraction of compute cost and can be omitted in practice when constructing a parameter-based estimate (see Table 1 in (Ivanov et al., 2021)). It is worth noting that the hardware-based estimation (see Section 3.1) includes these operations by default.

Numerical precision. AI models use various precision formats, from full-precision (FP32 - 32 bit floating-point) to lower-precision (FP16, FP8). Each halving of precision typically enables twice the operations per second on the same hardware. For example, using FP16 instead of FP32 can cut training time in half. However, when comparing models trained with the same total FLOP count (e.g., 1e26 FLOP), an FP32 model performs only marginally better than its FP16 counterpart. Given these minimal capability differences at equal FLOP budgets, developers often choose lower precision formats for their significant hardware efficiency benefits. The number format is only relevant when using hardware-based estimates, as the peak FLOP value needs to be representative for the training phase (see Section 4).

Quantisation. While post-training quantisation compresses model weights for inference efficiency and falls outside our cumulative compute definition, emerging training-time techniques like adaptive quantisation and binary neural networks fundamentally alter computation patterns. BitNet models replace matrix multiplications with binary operations and sparse additions, demonstrating energy reductions of 8.8x to 38.8x compared to traditional operations (Wang et al., 2023). This efficiency would significantly impact our parameter-based estimation formula, potentially reducing the coefficient from 6ND to as low as 0.16ND for large models[11]. When estimating cumulative compute for such architectures, we recommend calculating actual operations performed rather than theoretical full-precision equivalents to ensure consistent application of R1. As adaptive quantisation can have oscillating patterns (no, partial and full quantisation in a single training run), hardware-based estimation needs to use the respective numerical precision that is used during each phase.

Discarded branches. During development, it is common to run experiments to test ideas and verify scaling properties, without directly contributing to the parameters in the final trained model. Although these experimental branches consume significant compute resources, they are best viewed as contributing to the general advancement of the field rather than directly to an individual model's capabilities, similar to how insights from prior work are implicitly incorporated into new models. In line with R1, we recommend ignoring the compute spent on discarded branches and including only the compute spent on the final training run.

Multi-modality. Current frontier models are often multi-modal, meaning they can consume and produce text, images, and sometimes audio and potentially video in the future. This capability can be achieved through three approaches: (1) training separate models for each modality and fusing them together before release (Ho et al., 2022), (2) training all modalities together, or (3) using a partially frozen approach where some parts of an existing model are kept fixed while new capabilities are added (Llama Team, AI at Meta, 2024). Regardless of the approach taken to create multi-modal capabilities, at any point where the parameters of the final deployed model are directly updated should be accumulated together towards the compute threshold. For example, (1) would accumulate the individual models' training compute together, (2) only count the single model's training compute and (3) will need to account for some parameters not calculating gradients[12].

Model ensembles. Model ensembling leverages the fact that some models have different strengths and weaknesses—by smartly routing to the correct model (or filtering all models' responses by

[11] The standard formula for estimating neural network compute (6ND) can be approximately updated when using BitNet's 1-bit quantisation. Using energy consumption data from the BitNet paper for a 30B parameter model at 7nm, FP16 Transformer operations consume 7.66 picojoules (5.21 for MUL + 2.45 for ADD) while BitNet operations consume just 0.20 picojoules (0.06 for MUL + 0.14 for ADD). This represents a 38.3x efficiency improvement. Dividing the standard coefficient by this factor (6 ÷ 38.3) yields approximately 0.16, suggesting a modified formula of 0.16ND for estimating BitNet's inference compute requirements. The same approach can be done from a parameter perspective, creating "effective" parameters which will reduce the total parameters by 38 times.

[12] Frozen parameters of a model consume four FLOP instead of six during training for the parameter-based estimation, only losing the two operations during backpropagation—the weight gradient calculation and gradient accumulation over the minibatch size. This can be easily traced when performing forward and backward propagation (Bahdanau, 2022).

quality), their joint capabilities can improve compared to the individual participating model (Jiang et al., 2023). Our cumulative compute framework does not account for capability enhancements derived from non-parameter-changing techniques, and therefore would not aggregate compute across the constituent models. Importantly, ensembling typically does not produce emergent capabilities beyond those present in the constituent models, but rather optimizes the capability frontier already established by its components. Thus, if individual models in an ensemble do not possess high-impact capabilities, the ensemble itself is unlikely to manifest such capabilities, subject to the specific definition of high-impact in any given regulatory context.

Model merging. Model merging combines multiple models into a single model with enhanced capabilities, similar to but distinct from ensembling. Unlike ensembling, which maintains separate models, merging produces a new set of parameters, which qualifies for inclusion under R1. However, counting the full compute of all constituent models overestimates capability impacts, as merging techniques generally yield diminishing returns in capability enhancement relative to training compute invested. Currently, two distinct approaches to model merging exist: (1) averaging multiple checkpoints throughout a single training run to improve generalisation (Izmailov et al., 2018), and (2) combining models trained on entirely different data distributions to aggregate their distinct capabilities, sometimes called model soup (Biggs et al., 2024). We recommend accounting for unique compute cumulatively in both scenarios. In checkpoint averaging, this approach recognises that each new checkpoint (at time t) builds upon the previous checkpoint (at time $t-1$) with only incremental compute added—effectively degenerating into standard training with periodic averaging, a computationally inexpensive process already common in frontier model development (Llama Team, AI at Meta, 2024). For merging models trained on different data distributions, compute associated with any shared base model should be counted once, while the unique compute spent on specialising individual models should be cumulatively added after merging.

Synthetic data. Models can be trained on the output of other models, i.e., synthetic data. Our earlier case study on auxiliary models introduced synthetic data and the fact that its value is trivially transferable between training runs and does not directly update model parameters, acknowledging that it should not be counted in the definition of cumulative compute (R1). It also highlights the fact that it is similar to human-generated training data, which, even if we would want to account for it, it is difficult to audit a dataset to see if it contains synthetic data and guess what model generated it. However, this is to be tempered with the listing of "synthetic data generation" in recital 111 of the AI Act amongst techniques and methods that enhance capabilities of the model prior to deployment, and which should therefore count towards cumulative training compute.

Auxiliary models. There are many techniques that utilize more than one model during training, e.g., GANs (Goodfellow et al., 2014), model distillation (Gou et al., 2021), preference fine-tuning (Rafailov et al., 2023). In all of these cases, we recommend following our framework's R1 principle to determine whether compute should be counted. When auxiliary models generate non-reusable artifacts that depend on the specific model being trained (such as reference model log probabilities in DPO), this compute should be included in the cumulative total. For teacher/preference models that are frozen and only used for inference, only that inference compute should count. However, in cases like GANs, where a codependency exists between models (the generator actively improves the discriminator and vice versa), both training and inference compute should be counted as they fully satisfy our R1 criterion where the training signal is uniquely dependent on the current model state.

Bootstrapped models. The integration of pre-existing model parameters—whether through parameter initialisation or by using them as a starting point for mixture-of-experts architectures—presents computational attribution challenges. While these techniques satisfy our core criteria of directly affecting parameters (R1), they raise questions about how to aggregate compute. For example, should ten models, each trained with 1e25 FLOP, combined into a mixture-of-experts architecture be considered as having used 1e26 FLOP in total? Drawing a parallel to model ensembles, we can consider that bootstrapped models similarly leverage existing capabilities rather than necessarily creating new emergent ones. This suggests using the maximum compute of any

constituent model (rather than the sum) as the primary metric. However, this approach could potentially enable "hiding" FLOP by combining one large model with many slightly smaller models. While there is currently no evidence that this would become prevalent to circumvent compute thresholds or that it provides compute efficiency advantages, we recommend using the maximum compute approach as the default measurement for practicality. If evidence later suggests this is being exploited, regulators may need to reconsider accumulating all compute used across constituent models.

Changes to the training procedure. Training procedure changes are fundamental alterations to how AI models train, going beyond the standard steps of a single forward pass, backward pass, and gradient update. These modifications, exemplified by techniques like MCTS (Monte-Carlo Tree Search), can substantially enhance model capabilities through novel computational patterns (cf. Section 2.2). We propose evaluating such modifications through our framework on an individual basis. We caution against judging techniques from a particular outcome perspective, e.g., excluding safety-oriented training techniques from the definition of cumulative compute, as it may have unintended consequences. This is because it is often difficult to clearly distinguish between techniques that enhance safety versus capabilities, and safety techniques themselves can contribute to model capabilities. As a result, such exemptions weaken the coherence and utility of how we define cumulative compute.

Sparse models. There are two distinct types of sparsity patterns. Coarse-grained sparsity, as used in mixture-of-experts models, selectively activates only specific subnetworks during processing. Fine-grained sparsity, like NVIDIA's 2:4 pattern, systematically sets half of the parameters to zero in small chunks, enabling specialised hardware optimisations. Both approaches reduce computational costs during training and inference while maintaining model capabilities. When estimating training compute requirements (cf. Section 3), we should only count operations involving active parameters—in mixture-of-experts, this means counting only the activated expert computations, while for fine-grained sparsity patterns, we count only the operations involving non-zero parameters. For the parameter-based approach, this means adjusting N in the formula to reflect only the number of active parameters in each forward pass. The hardware-based approach accounts for coarse-grained sparsity naturally and fine-grained sparsity needs to substitute the according peak FLOP performance.

3 Measuring Training Compute

Estimating a model's training compute consumption can be achieved in two ways: tracking of GPU usage (hardware-based) and counting the expected FLOP through the model's architectural details (parameter-based).

This section provides a systematic analysis of these two approaches for quantifying training compute. We present the fundamental mechanics of each technique, conduct a comparative analysis of their technical characteristics, and evaluate their relative strengths for regulatory use. While these measurement techniques form essential building blocks for compute governance, their practical application will ultimately depend on how cumulative training compute is defined, which we discuss in Section 2.

3.1 Hardware-based Approach

The hardware-based approach involves accounting for the computational resources used during training. To estimate training compute, we need information on the number of GPUs used during training, the peak theoretical performance of the GPU (measured in FLOP/s) for a given numerical format (precision), the number of hours that training lasted and the GPU utilisation achieved (that is, the throughput achieved as percent of peak theoretical performance) (see Equation 1).

$$\underbrace{C}_{\text{Training compute}} = \underbrace{N}_{\text{GPU cluster size}} \times \underbrace{H}_{\text{GPU peak theoretical efficiency}} \times \underbrace{L}_{\text{Training duration}} \times \underbrace{U}_{\text{Achieved GPU utilisation}} \tag{1}$$

Current trends in AI infrastructure deployment demonstrate unprecedented scaling in centralised training clusters. Meta's Llama 3.1 405B utilised 16,384 H100 GPUs (Llama Team, AI at Meta, 2024), while GPT-4's training reportedly required approximately 25,000 A100 GPUs (Patel and Wong, 2023). The infrastructure trajectory points toward even larger deployments, with multiple companies planning clusters exceeding 100,000 GPUs by 2025 (Pilz et al., 2025b). xAI's Memphis cluster marked a significant milestone as the first installation to reach 100,000 H100 GPUs in July 2024 (Kan, 2024).

These massive training clusters are primarily useful for training operations, as inference workloads can be more efficiently distributed across smaller, geographically distributed hubs. While training runs may not always utilise the full cluster capacity, current training compute investment trends suggest labs will maximise compute for training operations. Recent deployments like DeepSeek V3's 2,048 H800 GPU cluster (DeepSeek-AI, 2024) represent optimisation strategies rather than a reversal of the scaling trend, as the primary computational demands remain concentrated in the training phase rather than deployment.

To put this into context, current state-of-the-art training runs rely on GPUs such as the NVIDIA H100, whose peak theoretical performance is close to 1e15 FLOP/s for the FP16 (16-bit floating-point) number format. There are dramatic differences between number formats, with lower bit-widths like FP8 typically offering two-fold improvement in performance compared to FP16. The choice of format for AI training is dependent on numerical stability, where FP16 is the current industry standard (at least publicly) and FP8 is showing promise (PyTorch, 2024, DeepSeek-AI, 2024). For hardware-based compute estimation, we only need to reference the GPU's peak theoretical performance for the specific number format used during training as listed in the hardware datasheets.

Training duration for modern state-of-the-art systems is usually within two to four months (EpochAI, 2025). While training duration has grown over time, it is currently believed that it is unlikely it will ever exceed a year-long duration (Sevilla et al., 2022a).

Model FLOP Utilisation (MFU) quantifies training efficiency by comparing actual processing speed to the theoretical peak performance. Specifically, MFU measures the ratio between achieved throughput

(tokens processed per second) and the maximum possible throughput given the model architecture and hardware specifications. In practice, language model training typically achieves MFU values between 20% and 60% (Sevilla et al., 2022b), reflecting fundamental computational constraints. During training, utilisation is limited both by how quickly the GPU can move data through its memory (akin to a traffic jam) and by the fact that some calculations need to happen in a specific order rather than all at once—meaning parts of the GPU simply have to wait for their turn to work.

A key advantage of hardware-based compute measurement is its technique-agnostic nature - it captures computational costs regardless of the specific training methodology, provided the computation occurs on hardware under direct control. This approach naturally accounts for complex training procedures like search-based algorithms (e.g., AlphaGo Zero's multiple forward passes for position evaluation) through direct hardware utilisation metrics. This can also become a disadvantage—if some techniques need special accounting through the cumulative compute definition, it is not possible to disentangle where specific FLOP went.

However, this measurement approach faces limitations when model training needs accounting for compute spent on inaccessible hardware, particularly relevant for techniques that use closed models behind an API if counted towards cumulative compute. This constraint highlights how practical measurement capabilities inherently influence the scope of what can be feasibly included in cumulative compute definitions.

Finally, implementing hardware-based compute verification requires comprehensive technical artifacts from training operations. Core requirements include GPU utilisation logs spanning the entire training duration, MFU metrics, and hardware specifications detailing peak FLOP at the utilised precision. While these artifacts primarily target pre-training measurement due to its current dominance in training compute costs, they must extend to any training phase that could consume significant compute under cumulative definitions.

This measurement framework faces additional complexity through mixed-precision training paradigms (Micikevicius et al., 2017). Modern training pipelines employ various numerical formats, each affecting peak FLOP calculations differently. While mixed-precision setups can be addressed by focusing on matrix multiplication FLOP (as FP32 element-wise operations are negligible, see Section 2.5), challenges emerge when multiple models share GPU resources with different precision requirements. This is further exacerbated with multi-tenancy setups, as the current technical reality is that GPU utilisation is not measurable for more than a single process[13].

Although such scenarios are likely rare in frontier training due to memory management complexities, the GPU utilisation metric does not differentiate between concurrent operations like FP16 training and FP8 inference on the same device. Given this technical constraint, we recommend adopting the smallest number format as a conservative baseline and using its respective peak FLOP—while this approach may overestimate total compute, it aligns with risk-based capability assessment objectives.

3.2 Parameter-based Approach

If one is privy to some details about model architecture and the planned training regime, it is possible to estimate the compute that will be spent training the model before training occurs. In particular, the cost of a forward pass for state-of-the-art models is roughly equal to twice the number of active model parameters (N)[14]. On the other hand, the cost of a backward pass is roughly twice that of a forward pass. Therefore, the cost of a single forward and backward pass is roughly thrice that of a forward pass, which costs 2N each, for a total of 6N.

[13] See https://forums.developer.nvidia.com/t/questions-on-per-process-gpu-utilization/265460/4.

[14] "Roughly" means that we are omitting FLOP stemming from the attention layer, which scales with the sequence length. If current training paradigms change and training happens primarily on very long sequence lengths (100K+), then more fine-grained formulas (like in the Appendix of (Narayanan et al., 2021)) need to be used.

Combining these facts, we can approximate the training compute (C) for a model as six times the product of the number of active parameters (N) and the number of training tokens (D) (see Equation 2). For context, labs have demonstrated the capability to train models at scales well beyond billions of parameters, processing dozens of trillions of tokens of text data.

$$\underbrace{C}_{\text{Training compute}} = 6 \times \underbrace{N}_{\text{Model parameters active per forward pass}} \times \underbrace{D}_{\text{Training examples}} \tag{2}$$

When using this approach, a key consideration is that some AI models fundamentally differ in how they utilise their parameters. In the simplest case, corresponding to models known as dense models, every parameter is active and used in each training step. However, sparse architectures like mixture-of-experts take a different approach by selectively activating only specific parts of the model. Therefore, only active parameters should be used for N in Equation 2 to account for this specific type of architecture.

There are also models that reuse parameters. For example, in computer vision, convolutional models repeatedly use the same parameters across different parts of an image, meaning their actual computational work depends heavily on the input image size and not just their parameter count. Similarly, attention mechanisms in language models have computational needs that scale with the length of the input text. This means that to accurately assess the computational requirements of these models, we need more than just the parameter count—we need to know the average text lengths in the training data.

While these considerations seem to complicate our calculations significantly, in practice, the simple compute estimation formula from Equation 2 works remarkably well. Let's look at a concrete example: One of the recently published models, Llama 3.1 405B, was primarily trained with moderate-length training examples of about 8,000 tokens, with only about 5% of its training involving very long texts (131,000 tokens). When we compare our simple estimation formula (3.8e25) with more detailed calculations that account for all these complexities (4.06e25), the difference is only about 6%—a small discrepancy given the scale at which these systems operate.

Given these empirical validations, we recommend using the straightforward 6ND formula for each training phase in most scenarios. However, when the hardware-based estimates deviate significantly from this formula, a more detailed calculation can become necessary—one that accounts for specific model architectures, input lengths, and training procedures.

The basic training compute calculation requires just two key metrics per training phase: active model parameters and total training tokens. For more comprehensive analyses, however, additional technical documentation becomes necessary.

First, system diagrams that map data flow across training phases, enabling accurate assessment under the cumulative compute definition. Second, detailed model architecture(s) specifying parameter utilisation patterns. Third, comprehensive training data statistics, including sequence lengths and token counts and similar information for other modalities.

Some training setups may involve additional compute-intensive elements, such as distillation processes, data generation systems, or other auxiliary models. As training methodologies evolve beyond current paradigms, documentation requirements will need to adapt accordingly.

Ideally, organisations conducting the training should provide mathematical formulas for each training phase and link them to the provided training documentation. This enables regulators to verify the calculations and assumptions. As a final validation step, we recommend cross-referencing these estimates with the hardware-based approach described earlier.

3.3 Comparing Both Approaches

Both approaches to measuring training compute have strengths and weaknesses.

Adaptability to the cumulative compute definition. The parameter-based approach enables precise computational attribution, mapping each FLOP to its training technique for threshold evaluation. Hardware-based measurement faces two key limitations: First, per-GPU granularity prevents separating utilisation between parallel processes, making it not suited if the definition requires tracking specific techniques. To address this, providers would need to dedicate separate hardware for tracked and non-tracked compute, potentially incurring communication penalties. We recommend that any cumulative compute definition avoid excluding specific techniques, as this creates implementation difficulties. Second, monitoring complexities complicate artifact submission—at 100K GPU scales, hardware failures occur approximately every 30 minutes, making individual GPU log submission unwieldy. For practical implementation, regulators should consider allowing aggregated metrics (average GPU participation, utilisation, and total runtime) rather than requiring exhaustive individual GPU logs.

Verifiability. Coarse-grained hardware-based estimates could potentially be verified through a combination of public information and cooperation from compute providers about providers' computing capabilities. In contrast, parameter-based estimates require architectural details and training dataset sizes, which developers may consider sensitive and proprietary information. While hardware utilisation rates might also be confidential, this information is typically less sensitive than architecture and dataset details.

Accuracy. When all required information is available, both methods produce similar estimates. Figure 4 compares these approaches using models from Epoch's database where providers disclosed the necessary details, showing that both methods yield comparable FLOP estimates. Even with some information needing to be guessed, (Sevilla et al., 2022b) found that both methods typically agree, and differ maximally within a factor of 2x.

Simplicity. Both techniques rely on multiplication of key information, making it trivial to perform. The parameter-based approach, when using the fine-grained estimate, introduces some complexity as each training phase needs to be completely understood to verify the final mathematical formula. Parts of these complexities can be alleviated with FLOP profilers[15] which provide model parameter utilisation patterns given a model architecture.

Feasibility of implementation. Training compute reporting is an unusual practice among developers (Sevilla et al., 2023). When it is reported, the method used to estimate it is not often commented on. Regardless, given the high stakes and high cost of compute, it is ensured that any major developer will keep careful accounting of its compute expenses and efficiency achieved, making these estimates possible and easy to perform internally.

Measurement in advance of training. The parameter-based estimate offers a theoretical framework that can be calculated before training begins, assuming the training proceeds according to plan. The hardware-based estimate can also be projected in advance, though it requires estimating expected GPU utilisation rates, which may vary due to technical factors such as hardware performance fluctuations during extended training runs[16]. Both approaches enable preliminary planning, which is particularly valuable given the substantial computational investments involved in large-scale training. Organisations typically develop detailed utilisation projections to optimise resource allocation, though real-world performance may deviate from theoretical estimates due to various technical and operational factors.

[15] See, for example, https://www.deepspeed.ai/tutorials/flops-profiler/#usage-outside-the-deepspeed-runtime.

[16] While GPU malfunctions can affect model training efficiency, their impact on compute estimates appears minimal (Erben and Erdil, 2024). For example, GPU failures every three hours during Llama 3 training only resulted in an estimated 1% reduction in Model FLOP utilisation.

Based on our comparative analysis, we primarily recommend hardware-based measurement for regulatory frameworks. This approach offers three crucial advantages: implementation simplicity, technique-agnostic nature, and preservation of proprietary training methodologies.

The effectiveness of hardware-based measurement depends critically on the granularity required by regulatory frameworks. When operating at the training phase level with aggregate accounting, this approach enables straightforward verification. However, for technique-specific requirements, only parameter-based measurement can provide sufficient granularity.

This relationship between the definition of cumulative compute and measurement methodology has far-reaching implications for the compute threshold regulation. The regulatory specification of cumulative compute, if done haphazardly, may not only influence how computation is measured, but also how training architectures evolve and how organisations structure their development pipelines.

Finally, we want to stress the fact that these techniques do not provide the factually correct amount of compute that has been used during training, only an approximation. This fundamental limitation stems from various technical factors, including GPU driver configurations, compilation optimisations, and hardware instruction sets, which can all impact the final FLOP calculations. Consequently, any governance framework for compute measurement would need to acknowledge these inherent approximations while establishing reporting standards for model developers that balance accuracy with practical implementation constraints, for example through an allowed margin of error.

4 Setting the Cumulative Compute Threshold

We have made the case earlier in this report that computing thresholds should aim to reliably identify frontier AI capabilities while minimising regulatory burden on less capable systems. While training compute scaling continues to drive capability advances (Denain, 2024), algorithmic efficiency improvements have progressively lowered the training compute required to achieve a given level of capabilities (Ho et al., 2024), with DeepSeek-v3 and -R1 models being the latest exemplification of this trend (DeepSeek-AI, 2024). This dynamic creates a core challenge for threshold adjustment: identifying the most advanced models that warrants oversight while maintaining a focused regulatory scope.

This analysis attempts to present a practical approach to setting compute thresholds that effectively identify frontier AI capabilities while minimising unnecessary regulatory burden. Our goal is to provide a structured framework balancing oversight effectiveness, implementation feasibility, and market dynamics to guide threshold calibration decisions. By examining these key factors, regulators can make informed adjustments that maintain governance objectives without compromising innovation. Let's now explore the fundamental regulatory design parameters that shape threshold decisions.

4.1 Regulatory Design Parameters

Our view is that the primary function of compute thresholds should be to identify models that could represent significant advances in AI capabilities. Current industry trends indicate accelerating training compute investment, with 31 confirmed computing clusters under construction for 2025 releases (Pilz et al., 2025b). This acceleration suggests that the predicted trend of training compute growth on model training is likely to continue.

Regulatory capacity presents a critical operational boundary for threshold adjustment. With aggressive projections indicating hundreds of models potentially exceeding the 1e25 FLOP threshold by 2026, the enforcement authorities could face significant bandwidth limitations (Figure 6). Each conformity assessment may require substantial technical expertise and resources, necessitating careful consideration of assessment scope and frequency.

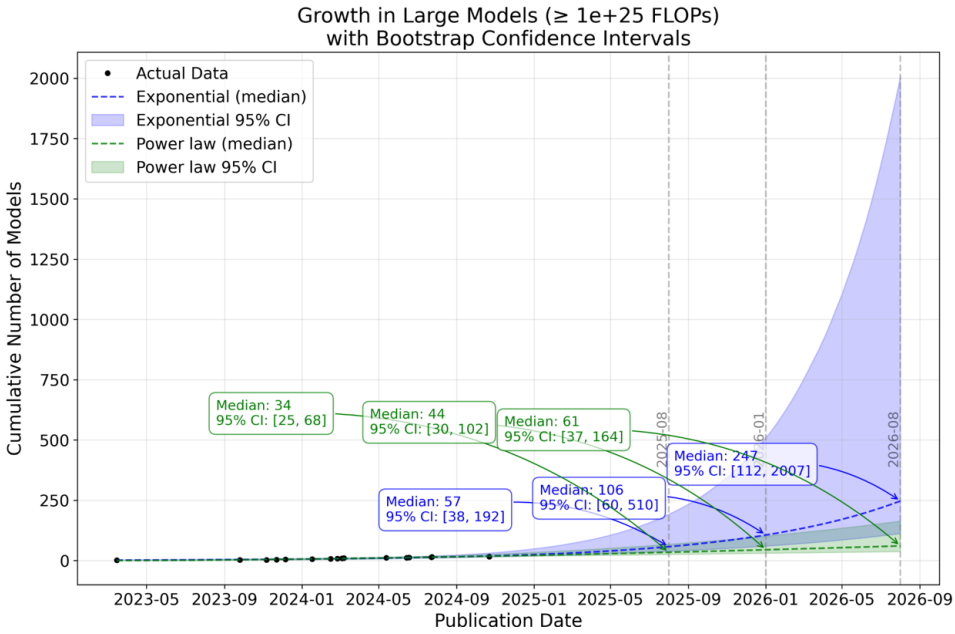
The selection of compute thresholds may have significant implications for global AI development. Conservative thresholds that capture a broader range of models could impede market access and innovation, particularly affecting research institutions (when putting AI models to use outside of research) and smaller developers. Conversely, liberal thresholds might compromise effective oversight of potentially transformative systems.

The determination of appropriate thresholds must account for both immediate oversight requirements and longer-term strategic implications for the EU AI ecosystem. This creates a fundamental trade-off between maintaining robust governance and fostering technological advancement within the European market.

An effective compute threshold framework would identify a select number of frontier models annually, enabling focused risk assessment and development of corresponding safety measures. Under current scaling trajectories, this would suggest regular threshold increases aligned with the four- to five-fold annual training compute growth. However, two critical failure modes could undermine this approach:

— First, a shift toward applying less compute-intensive post-training techniques to static base models could decouple the compute-capability relationship. In this scenario, organisations would reuse existing pre-trained models and enhance their capabilities through compute-efficient post-training methods, without incurring significant additional compute costs. This would render compute thresholds ineffective as a preventive governance tool, as

Figure 6: Prediction of cumulative number of models >1e25 FLOP for three future dates (August 2025, January 2026 and August 2026). The large confidence intervals show the uncertainty in this prediction, ranging from 25 models to 2007 at the lowest and highest points of both functions.



Source: Own elaboration with data from EpochAI.

capabilities could substantially advance while maintaining nearly the same total compute expenditure. Attempting to capture such advances by dramatically lowering thresholds would flood the market with regulatory oversight requirements while failing to reliably identify frontier capabilities.

— Second, accelerating algorithmic efficiency gains could outpace training compute scaling, requiring downward threshold adjustments. This could necessitate additional constraints (e.g., development date requirements) to avoid capturing legacy systems, significantly complicating the regulatory framework while potentially missing frontier capabilities.

While the first failure mode would invalidate compute-based governance entirely, current evidence shows no signs of this occurring (cf. Figure 1). The second mode, rapid algorithmic efficiency gains, presents a more pressing challenge for threshold calibration. Obtaining empirical evidence on algorithmic efficiency at frontier scales is prohibitively expensive, creating significant gaps in our understanding of current capabilities (Ho et al., 2024). Moreover, algorithmic breakthroughs tend to occur suddenly rather than incrementally, further complicating regulatory responses. Our proposed adjustment framework accounts for these uncertainties, providing a systematic approach for threshold calibration as technical capabilities evolve.

4.2 Adjustment Framework

Our approach builds on several foundational observations about the current state of frontier AI development:

— Training compute scaling continues to outpace algorithmic efficiency gains (though measurement uncertainty remains high).

— Development of frontier models remains concentrated among a small set of well-resourced organisations.

— Assessment capacity within regulatory bodies faces clear operational constraints.

Given this situation, we propose an initial framework that prioritises regulatory feasibility while maintaining meaningful oversight of frontier capabilities. It is anchored in three primary regulatory objectives:

— **Frontier Model Coverage**. Current AI development remains concentrated among a small set of organisations with sufficient computational resources. This creates a natural target for regulatory oversight: capturing the top-N most computationally intensive models within each assessment period. The specific number of models needs to be determined by a market analysis of current frontier labs.

— **Assessment Period Stability**. The threshold must remain effective throughout the designated review period. Given observed training compute scaling rates between four- to five-fold annually, this requires forward-projection of model development trajectories. For example, a 6-month review period would necessitate setting thresholds to anticipate two-fold training compute growth to maintain consistent coverage of frontier models.

— **Regulatory Resource Alignment**. The enforcement authority would need to project the necessary resources to capture the number of models required to track the capability frontier.

Beyond these primary objectives, the framework must account for several secondary objectives that influence threshold adjustment decisions:

— **Risk Verification Requirements**. The effectiveness of compute thresholds must be validated through systematic risk assessment of captured models. When evaluations consistently demonstrate minimal risk profiles (potentially through current risk management approaches (Anthropic, 2023)) signals that thresholds may be set too conservatively and should be adjusted upwards to avoid unnecessary regulatory burden. Conversely, for models showing significant risks, these must be rigorously documented through empirical analysis including post-deployment monitoring and capability assessments (Bernardi et al., 2024).

— **Capability Diffusion Management**. The framework explicitly acknowledges that capabilities initially achieved at frontier compute scales will likely become accessible at lower compute requirements over time. Rather than responding to this diffusion by lowering thresholds—which could capture numerous legacy systems—the primary response should leverage alternative regulatory mechanisms. Threshold reduction should serve as a last resort when other oversight tools prove insufficient.

— **Market Impact Assessment**. Before adjusting the threshold, regulators should analyse the broader market implications beyond direct model coverage. Any threshold modification should consider developer planning horizons, as training runs typically take 2-6 months. With bi-annual reviews, regulators should publish forward-looking guidance that enables providers to align development roadmaps, avoiding mid-development regulatory surprises unless extraordinary circumstances warrant emergency adjustments[17]. This predictability creates a stable environment for innovation while maintaining effective oversight.

[17]To create effective forward-looking guidance, regulators could publish projected threshold ranges based on observed compute scaling trends (e.g., 4-5x annual growth), allowing developers to anticipate future requirements. These projections would be non-binding but would signal the expected trajectory absent significant technological disruptions, giving providers a planning horizon that aligns with typical 2-6 month training cycles while preserving regulatory flexibility to respond to unexpected breakthroughs.

4.3 Implementation Approaches

We have identified three approaches to implementing threshold adjustments, each offering distinct advantages and challenges for maintaining effective regulatory oversight:

- **Dynamic thresholds**. This approach links adjustments directly to the scale of frontier AI models—for example, setting thresholds one order of magnitude below the largest known training runs. While this method provides automatic calibration to technological progress, it introduces significant limitations. First, it may create planning uncertainty for model developers. Second, it could incentivise strategic information disclosure from leading AI laboratories to influence threshold levels. Most critically, this approach may not adequately capture improvements in algorithmic efficiency that enable similar capabilities with reduced compute requirements.
- **Regular review periods**. Under this system, thresholds would be reassessed at predetermined intervals—perhaps annually or bi-annually—through a structured process involving consultation with industry and technical experts. This approach enables thorough consideration of technical advances, empirical risk evidence, and regulatory capacity constraints. However, the predictability of review cycles may lead to strategic timing of development schedules by providers and strategic information disclosure to influence threshold levels. Additionally, fixed review periods may prove insufficiently responsive to the rapid pace of technological advancement in AI development.
- **Ongoing assessment**. Delegation to expert bodies, such as scientific panels or dedicated regulatory offices, offers a more flexible approach. This mechanism allows continuous evaluation of technological developments, emerging risks, and regulatory effectiveness. While this approach benefits from dedicated technical expertise and enhanced responsiveness, it requires substantial institutional capacity and robust governance frameworks to maintain transparency and accountability. The primary challenge lies in managing market uncertainty, as providers may face unexpected conformity assessments following threshold adjustments. Additionally, this approach creates vulnerability to strategic disclosure, where providers could manipulate assessment timing and threshold adjustments to gain competitive advantages without actually deploying the disclosed capabilities.

Each of these approaches could be combined with supplementary mechanisms. For instance, emergency adjustment procedures could allow for rapid responses to unexpected technological breakthroughs, while mandatory minimum periods between adjustments could provide stability.

Tentative Recommendation. Based on our analysis of implementation approaches, we gravitate towards ongoing assessment through a dedicated regulatory body. This recommendation stems from several technical and institutional factors.

First, this approach enables continuous evaluation of the relationship between training compute scale and capabilities, rather than relying on point-wise assessments that may miss important trends or emerging risks. The continuous nature of the assessment allows for building institutional expertise in tracking compute utilisation patterns, understanding their implications for model capabilities, and maintaining close communication with technical stakeholders.

Second, a key advantage of this mechanism is the development of deep technical competency within the regulatory body. Rather than convening periodic committees, ongoing assessment requires maintaining in-house resources with sustained expertise in both the technical and regulatory aspects of training compute measurement. This expertise is crucial for evaluating how emerging capabilities and training compute efficiency gains interact with existing threshold levels.

While this approach introduces some market uncertainty compared to fixed review periods, the benefits of responsive oversight likely outweigh these limitations. The approach allows for rapid

adaptation to technological breakthroughs while maintaining consistent evaluation criteria. Moreover, the presence of a permanent regulatory body facilitates international coordination and information sharing, which is essential for preventing regulatory arbitrage.

The implementation of this approach requires careful consideration of institutional design and procedural safeguards. Should technological advances fundamentally alter the relationship between training compute and capabilities, the effectiveness of this approach would need to be reevaluated. However, given the novel nature of compute-based regulation, the flexibility and technical depth enabled by ongoing assessment currently appears most appropriate for this regulatory challenge.

5 Notifying of Compute Threshold Exceedances

The EU AI Act establishes notification requirements for GPAI models that exceed or will exceed the compute threshold. Article 52(1) states that providers must notify the Commission when their model meets the compute threshold for presumption of systemic risk, or within two weeks after it becomes known that this will be the case. This raises an important question: at what point in the development process can a provider reasonably be expected to "know" that a model will exceed the threshold?

How exactly AI models are developed is not transparent to external observers. While the specific development processes vary across organisations and remain largely proprietary, we can make reasonable inferences based on fundamental economic principles—when organisations invest significant resources in compute (and therefore money), they necessarily engage in substantial planning.

The model provider gains increasing clarity about compute usage the closer they get to the final training run, creating a trade-off: earlier notification provides regulators more time but comes with greater uncertainty, while later notification offers higher certainty but less time for review. Furthermore, depending on what information must be reported, providers may be unable to supply complete documentation at earlier stages of development. Below we list some potential notification points along the development timeline.

5.1 Knowledge Points in Model Development

The development of current frontier AI models involves multiple stages where knowledge about final compute usage becomes increasingly definitive[18]. Each potential notification point offers different trade-offs between regulatory preparedness and implementation practicality. While we present typical timelines below, these should be understood as approximations that can vary considerably across organisations.

As training methodologies evolve, these discrete knowledge points may become less clearly defined. Several emerging paradigms could significantly alter when providers can reasonably "know" they will exceed compute thresholds:

- **Continuous Training Paradigms**: Providers could be moving toward persistent training where pre-trained models undergo continuous updating rather than discrete training runs. In this approach, compute accumulates gradually over extended periods without clear start and end points.
- **Large-scale post-training**: When providers significantly expand existing models with additional compute—through training on vast amounts of synthetic data, reinforcement learning from human feedback (RLHF), or other post-training enhancements—the temporal relationship between development phases can compress substantially. This compression occurs as providers can reuse the existing architecture, eliminating the need for architectural design. Additionally, specific training techniques may require fewer scaling experiments, further reducing the end-to-end time until training completion.

Therefore, any notification requirement should be adaptable to changing timelines between knowledge points, or instances where certain development phases become unnecessary altogether while maintaining focus on the core question of when a provider has sufficient certainty about exceeding the threshold.

[18]The knowledge points presented here largely reflect current development lifecycles and may not hold true as the field evolves. For instance, scaling experiments could become obsolete with improved theoretical understanding of scaling laws, or when model capabilities advance through mechanisms other than increased scale.

Table 2: Knowledge points in AI model development and their implications for notification timing.

Knowledge Point	Timeline Relative to Training	Key Information Available	Notification Pros	Notification Cons
1. Infrastructure Planning	3-12 months before training begins	Architectural design concepts, approximate parameter count, maximum compute availability	Earliest possible notification, maximum preparation time	Very high uncertainty
2. Architectural Design	1-3 months before training begins	Model architecture, rough parameter count, amount of training data, preliminary training plan	Early notice with basic specification details	Still substantial uncertainty in compute estimates
3. Scaling Experiments	2-6 weeks before training begins	Hyperparameters, validated architecture	More reliable computing estimates based on empirical data	Some uncertainty remains, methodology may evolve
4. Pre-training Resource Commitment	Days to 2 weeks before training begins	Finalised configuration, committed resources, training pipeline	High-confidence estimates before major compute expenditure	Limited preparation time for regulators
5. Training Start	Training day 0	Initial compute consumption metrics, actual resource utilisation	Concrete implementation begins, direct measurement possible	After significant resources committed, late for prevention
6. Training Completion	After full training run (typically 2-6 months)	Final compute usage with complete certainty	Perfect accuracy in reporting	Too late for preventive governance

Source: Own elaboration.

Having mapped these key knowledge points in the development timeline, we can now identify what we consider to be the most practical notification trigger point. This selection must balance estimation certainty against the regulatory goal of receiving notification before major compute expenditure occurs.

5.2 Practical Implementation and Recommendations

Based on the knowledge points outlined above, notification at the pre-training resource commitment stage (point 4) represents the most balanced approach for current model development methodologies. At this stage, providers have high-confidence estimates and committed resources, but have not yet initiated the primary compute expenditure. This timing satisfies Article 52(1) by providing notification at the earliest practical point when knowledge becomes reasonably certain.

The viability of this trigger point rests on the premise of discrete training runs. Although this remains the standard, it may prove insufficient in the future as alternative training paradigms gain traction. Continuous training regimes, where models are persistently updated, lack a singular commitment moment. In such cases, notification may be better tied to the point at which the provider commits to a training regime projected to exceed the compute threshold within a defined time-frame. Similarly,

in cross-provider training scenarios—such as large-scale post-training—notification should occur when it becomes reasonably certain that cumulative compute (original training plus enhancement) will exceed the threshold.

Article 52(2) permits providers to demonstrate that a model does not pose systemic risk despite exceeding the compute threshold. We note that this demonstration may be difficult to execute at the pre-training resource commitment stage, when the model is not yet instantiated. Nevertheless, providers could, amongst other things, reference architectural or functional comparisons to existing models.

According to Article 52(1), the notification should include evidence that the compute threshold has or is expected to be met. We recommend that the evidence provided enables the competent authority to validate the provider's compute estimation that has triggered the obligation to notify. At our proposed notification point 4, providers would have completed preliminary scaling experiments and would possess the following information:

1. Detailed compute capacity allocation, including cluster specifications for each training phase (e.g., GPU/TPU count, hardware specification, potentially networking specification in case of geo-distributed training).
2. Expected model efficiency metrics (e.g., model FLOP utilisation rates based on scaling experiments for each training phase, employed numerical precision).
3. Compute estimation methodology with underlying assumptions (e.g., ablation experiment artifacts, model architecture, training recipe, dataset size, estimated hardware failure rates).
4. Planned minimum and maximum training duration scenarios.
5. Expected training timeline with key milestones and anticipated completion date.

Section 3.3 ("Infrastructure, Scaling, and Efficiency") and 3.4 ("Training Recipe") in (Llama Team, AI at Meta, 2024) exemplify this technical documentation, though they lack complete training dataset information, use measured rather than estimated hardware failure rates, and omit specific training timelines. Ultimately, the complexity of this documentation should be proportionate to the verification needs and risk assessment goals[19].

For regulators developing implementation guidance, we recommend creating guidelines that:

1. Focus on the substantive question of "when it becomes known" rather than assuming specific development methodologies.
2. Clarify notification rules for both discrete training runs and emerging training paradigms, such as large-scale post-training and continuous training regimes.
3. Provide clear standards for what constitutes "reasonable certainty" about threshold exceedance.

This approach attempts to preserve the preventive governance objectives of the AI Act while accommodating technological evolution in AI development practices.

[19]For example prediction evaluations, as outlined in (Chan, 2024), enable regulators to assess whether providers truly understand how their models will behave before full training begins. By requiring providers to predict test evaluation results based on initial evaluations, the competent authority can evaluate providers' understanding of model generalisation and potentially impose additional requirements if their predictive capability is insufficient.

6 Verifying Training Compute

The AI Act includes two conditions under which providers of GPAI models must document or report information related to the amount of computational resources used to train their model.

Firstly, under Article 52(1), if a GPAI model meets the compute threshold or is expected to do so, its provider must notify the Commission within two weeks, providing "the information necessary to demonstrate" that these criteria were or will be met. Hence, if a model crosses the compute threshold, its provider must demonstrate this to the regulators by disclosing information on training compute usage.

More generally, and independently of whether a model meets the conditions for being classified as a GPAI model with systemic risk, Article 53(1)(a) mandates that all GPAI model providers must draw up technical documentation of their models and provide relevant information from this documentation to the AI Office or national competent authorities upon request. In particular, based on Annex XI(c), this documentation must include "the computational resources used to train the model (e.g., number of floating point operations), training time, and other relevant details related to the training", as well as "design specifications of the model and (the model's) training process" and "known or estimated energy consumption of the model".

The enforcement authority could verify compliance with these reporting and transparency requirements, so that if appropriate, it could proactively classify potentially non-compliant GPAI models as GPAI models with systemic risk according to its competencies in Article 52(1).

Two primary concerns emerge in the context of non-compliance:

1. **Under-reporting of training compute**. GPAI providers may under-report the amount of training compute used to avoid obligations associated with being classified as a provider of GPAI models with systemic risk.
2. **Non-declaring of models**. GPAI providers may entirely fail to report large training runs, thereby circumventing all regulatory obligations.

In this section, we discuss several verification methods that regulators could use to address these concerns. These methods vary in terms of implementation feasibility, level of assurance, and intrusiveness. Some methods are low-touch and implementable with almost no additional effort beyond enforcement of existing obligations, while others would require significant international cooperation and technological innovation. The purpose of this discussion is not to suggest a particular verification regime, but to give an overview of the technological option space, and of legal and practical constraints that emerge in the context.

6.1 Verifying Declared Amounts of Training Compute

We present two methods to verify reported training compute: one method that focuses on comparing different measurement approaches, and another method that verifies reports by obtaining information from cloud service providers (CSP).

Concordance between measurement approaches. The regulator could verify the amount of training compute reported by a provider by comparing FLOP estimates from multiple measurement approaches. The more approaches are compared, and the more concordance there is between the different approaches, the harder under-reporting becomes. With the dual estimation methodology discussed earlier, the enforcement authority would already require the provider to report their training FLOP through two approaches—one based on hardware, and one based on parameters. If these estimates are incongruent to an implausible degree (i.e., off by >10x), or do not match other

training-related information disclosed by the providers, then this would be evidence of non-compliance. For further due diligence, the regulator could request a more detailed justification for each estimate, including disclosure of the input factors to the equations discussed in Sections 3.1 and 3.2.

The enforcement authority could also derive its own FLOP estimate based on a method different from the provider's and use this for comparison. For example, it could compare the provider's hardware- and parameter-based compute estimates with a compute estimate based on hardware usage hours inferred from energy consumption during model training. The data required for this verification approach would have to be sufficiently granular, and could be derived either on the basis of the reporting requirements in Annex XI(2)(e), or from wider corporate reporting requirements on the environmental impact of model training.

Implementation of this verification approach appears to be highly feasible. Due to the comprehensive transparency obligations in Article 53(1), as well as the power to request documentation and information enshrined in Article 91(1), the regulator could request the information required for different measurement approaches. One weakness of this approach is that it would not allow for detection of fraudulent reporting. The information disclosed by providers can only be verified via triangulation, as direct, un-intermediated verification would require hardware access by the enforcement authority or a trusted third party. Triangulating between different providers could increase reliability, but would still not provide complete assurance. Moreover, hardware-based and parameter-based compute estimates can differ by up to a factor of 2 (Sevilla et al., 2022b), which makes it harder to attribute incongruence to fraudulence.

Workload verification through cloud service providers (CSPs). The regulator could verify the amount of training compute reported by the provider by comparing it to information obtained from CSPs (Heim et al., 2024). Large-scale training runs are conducted in data centres containing tens of thousands of AI accelerators, which GPAI providers access through an Infrastructure as a Service (IaaS) business model offered by CSPs like Amazon Web Services, Microsoft Azure, and Google Cloud. For business purposes like billing and performance monitoring, these CSPs have access to a wide range of data on GPAI providers' computational workloads. Verification of training run size would require that cloud compute providers inform the regulator about the type and magnitude of these workloads:

- **Type of workload**. Allowing the regulator to distinguish large-scale GPAI training runs from other AI workloads, like design, enhancement, and deployment, as well as non-AI workloads, like scientific computing. A simple indicator of workload type is the hardware configuration available to a customer—for example, if a customer has access to tens of thousands of accelerators that are connected via a high-bandwidth network, and if this hardware was used for a certain length of time, this would be evidence that a large-scale training run was conducted. Information on hardware configuration could be obtained on the basis of billing-related technical information already conducted by the compute provider. More precise classification could be achieved on the basis of cluster-level, or node-level workload features, which could be either defined manually, or derived through a machine learning classifier.
- **Magnitude of workload**. Allowing the regulator to quantify the amount of compute available to a GPAI provider on a rented cluster ('compute budget'), as well as to determine how much of it is allocated to GPAI training runs. The compute budget can be estimated either theoretically, by multiplying the summed theoretical peak performance of all available AI accelerators with available usage time and assumed average utilisation; or empirically, using metadata on cluster activity, such as on data traffic within and between accelerators and nodes. Although the empirical approach is more precise than the theoretical approach, both approaches merely provide an upper bound to training run size for a given customer. More precise measurement would require a combination of approaches, and would benefit from including a wide range of high-level and low-level workload measurements.

A detailed list of observable data attributes that the regulator could require CSPs to provide for this purpose is given in Appendix B of (Heim et al., 2024).

Accessing data from CSPs would allow regulators to verify training run size directly, without having to rely on reports by GPAI providers, although this advantage may be less valuable if GPAI providers and CSPs are closely integrated, or even part of the same company. Moreover, despite the direct hardware access, sufficiently incentivised providers might be able to evade oversight by splitting training workloads across multiple compute providers, or by not reporting training runs in the first place.

Technical implementation of this approach would be feasible, as CSPs already gather much of the required data, and have the expertise to develop technological solutions for advanced workload classification and quantification. Legal implementation seems more challenging. Even if it was possible, around 75% of data centres are located outside of Europe (Pilz et al., 2025a), so verification would have to rely either on extraterritorial enforcement of EU law or on coordination and partnership with other relevant jurisdictions, such as the United States. Given these implementation challenges, it is worth considering a voluntary approach, in which providers are encouraged to request CSP data as a proof of compliance, but CSP's are not legally obligated to comply with these requests.

6.2 Verifying That No Training Runs Are Undeclared

Assurance that no training runs are undeclared requires identifying actors with access to large-scale computational infrastructure and ensuring that these actors have not deployed models in the EU market without having notified the enforcement authority. We discuss two approaches to how such identification could be achieved.

Identifying large customers of cloud providers (KYC). Regulators could require cloud compute providers to identify entities with access to amounts of compute above a certain threshold. Depending on the purpose and scope of such a scheme, identity information could include details on the corporate structure of the entity, its beneficial owners, the purpose for the use of compute, and whether it matches particular risk criteria. The regulator could access this information for ad-hoc compliance checks if it suspects that an entity is deploying models illegally in the EU, or define requirements for proactive reporting of such cases. (Egan and Heim, 2023) outline a detailed proposal for such a KYC scheme in the US regulatory context.

Analogous schemes already exist in the financial sector to prevent money laundering and terrorist financing, for example, in the EU "Anti-Money Laundering Package" of directives and regulations (TaylorWessing, 2024). In the context of AI, a basic version of a KYC scheme was mandated by US Executive Order 14110, which required US cloud service providers to identify any foreign person that is training a highly capable AI model using their infrastructure, and to report any instances of such training to the US Secretary of Commerce (Executive Office of the President, 2023). Like workload verification, a comprehensive KYC scheme would require extraterritorial enforcement of EU legislation or international cooperation, as most data centres are located outside the EU.

Detection of large compute clusters. Regulators could create and maintain a map of known compute clusters, including information about their size and their owners. Companies known to be in possession of large clusters would be likely candidates for training or hosting the training of models above the regulatory threshold. This process could be done using publicly available information, disclosing requirements, or unilateral monitoring.

For the first method, regulators could make use of public reports on companies' computational resources, for example, Meta publicly announced the acquisition of 350,000 H100 GPUs (Lee et al., 2024). For the second method, regulators could follow a model similar to section 4.2(ii) of US Executive Order 14110, which requires companies to report the existence and location of compute clusters exceeding 1e20 FLOP/s theoretical maximum computing capacity (Executive Office of the President, 2023). This could be complemented by requiring chip vendors to disclose large-scale GPU

transactions, helping create a more complete map of compute ownership. Finally, undeclared clusters could be detected through unilateral monitoring, for example using satellite images, remote sensing, energy tracking, and other forms of intelligence (Wasil et al., 2024, Baker, 2023).

Implementation of a detection scheme is facilitated by the fact that, at least given the current state of technology, only relatively few actors have access to compute clusters for GPAI training runs above the compute threshold, and that GPU clusters are fairly geographically concentrated. However, monitoring schemes involving compute providers and chip flows would face similar issues related to jurisdictional reach as described in earlier sections: only the minority of compute clusters and chip vendors is located in the EU. Concerns about intrusiveness could arise, especially from unilateral monitoring, so this approach may be most appropriate as a last resort to target high-risk actors under suspicion of persistent fraudulent reporting.

6.3 Verification of Both Declared and Undeclared Training Runs

A number measures are useful for both verifying declared training runs, and for ensuring that no training runs are left undeclared.

Verification based on observed performance. Scaling laws predict that language model test loss (performance of a model on unseen data) decreases smoothly with increases in model size, data set size, and pre-training compute. (Kaplan et al., 2020) test loss, in turn, can be used to predict model performance, as described by (Owen, 2024). In reverse order, this relationship could be used to infer pre-training compute based on benchmark performance, without having to rely on sensitive technical information. In theory, this method could be used both for verifying reported training runs, and for detecting undeclared training runs. For example, it could be used if regulators want to conduct spot checks of whether compute for models clearly below the threshold was reported correctly.

Whistleblower protection. A whistleblower protection program would allow employees of GPAI providers and CSPs to speak up safely about compliance violations related to reporting requirements for GPAI providers, or obligations in the EU AI Act more broadly. Employees may submit reports anonymously through secure communication channels, and those who choose to identify themselves could be guaranteed confidentiality and protection from any form of retaliation. Article 87 of the EU AI Act already emphasises that EU Directive 2019/1937—which aims to "guarantee a high level of balanced and effective protection for persons reporting on breaches of (EU) rules" (European Commission, 2019)—shall apply. The enforcement authority could determine whether this directive is already sufficient, or if additional measures are warranted for the specific purpose of enforcing EU AI Act obligations on GPAI providers.

End-to-end verification. Several hardware-based solutions for high-assurance GPU tracking and AI workload classification and quantification were proposed. For example, (Shavit, 2023) proposes a hardware-enabled monitoring system for machine learning workloads, which involves on-chip activity logging, regular inspection of logs by regulators, and chip supply chain monitoring. If implemented comprehensively, such a system could be highly effective at detecting various types of non-compliance, including under-reporting of training compute and undeclared training runs. However, the required technology is not available yet, and comprehensive implementation would require a concerted global effort by a large number of stakeholders, including governments, chipmakers, CSPs and GPAI providers.

7 Conclusions

Drawing from expert critiques and recent research, this section examines fundamental challenges in using compute-based thresholds for AI regulation. We analyse both technical and practical limitations of current approaches while exploring alternative metrics, ultimately providing recommendations for making compute thresholds a more robust regulatory tool.

7.1 Compute Threshold Limitations

(Hooker, 2024) critiques the use of compute thresholds to regulate AI models and systems, arguing that this approach oversimplifies a complex challenge and may prove ineffective at mitigating genuine risks. While we acknowledge the validity of several concerns raised, we examine whether our proposed framework resolves these identified limitations. Below, we address Hooker's key objections by demonstrating how our methodology responds to these concerns.

Relationship Uncertainty: Hooker argues that the correlation between compute and model capabilities is increasingly unpredictable, with smaller models often outperforming larger ones. Answer: We acknowledge that algorithmic efficiency improvements will enable smaller models to achieve capabilities previously requiring more compute. However, Article 51(1)(b) provides an additional way of designating a model to be GPAI with systemic risk based on their capabilities and impact. This means that we do not need to lower the thresholds when systemic risks affect smaller models, while still capturing the cutting-edge capabilities that emerge from the largest compute expenditures. The correlation of compute and capabilities is discussed in depth in Sections 2.1 and 2.2.

Arbitrary Thresholds: Neither the US Executive Order's 1e26 FLOP threshold nor the EU AI Act's 1e25 FLOP threshold has clear scientific justification. Answer: We agree that transparent reasoning and clear communication about potential future changes are essential for both regulatory effectiveness and industry compliance. Our approach to using compute thresholds to identify the top-N models is straightforward to implement, and we provide the scientific justification for this approach in Section 4.

Measurement Problems: Current policies fail to specify how to handle technical complexities like quantisation and modern architectures. Answer: While these are valid concerns, we propose a two-track approach to standardisation. Hardware-based estimation offers a reliable foundation that remains consistent regardless of model architecture or quantisation choices. When using parameter-based estimation techniques, we outline standardised ways to deal with complexities in Section 2.5. We acknowledge however that maintaining measurement integrity requires ongoing updates to these methodologies as AI architectures evolve, particularly to prevent gaming of metrics.

Focus on Training Compute: By focusing solely on training compute, these thresholds miss crucial factors like optimisation techniques and data quality that often drive capabilities. Answer: While data quality significantly impacts model capabilities, improvements in the area that enable enhanced capabilities with reduced training compute should be reflected in threshold adjustments, as detailed in our Section 4.

Static Nature: Fixed thresholds may quickly become obsolete as AI technology rapidly evolves. Answer: Given this rapid pace of AI advancement, we propose an ongoing assessment strategy rather than fixed intervals for threshold calibration, with detailed methodology outlined in Section 4.3.

7.2 Inference Compute Scaling

Advanced inference optimisation techniques, including chain-of-thought prompting, best-of-N sampling, and Monte Carlo Tree Search -collectively termed "test-time compute"—enable enhanced

model performance through increased computational investment during inference (Villalobos and Atkinson, 2023). This methodological innovation introduces significant implications for the traditional relationship between training compute and model capabilities. Specifically, these approaches suggest that advanced capabilities might be achievable with relatively smaller-scale models, contingent upon increased allocation of computational resources during inference.

Ultimately, since these inference techniques would be adopted downstream, we consider them outside the purview of compute thresholds aimed at model developers. This does not mean that unifying training compute with downstream inference compute for threshold generation could not be useful, depending on the context."

However, it is crucial to note that these optimisation techniques can be integrated into the training pipeline itself. In such cases, the additional computational costs should be incorporated into total training compute estimates, as detailed in our analysis of "Changes to the training procedure" in Section 2.5 and the Section 2.3.

7.3 Alternative Metrics to Cumulative Compute

Annex XIII of the AI Act establishes guidelines for identifying alternative metrics to determine GPAI systemic risk beyond raw computational resources. Among these alternatives, standardised benchmark performance thresholds offer a direct capability assessment.

Training cost thresholds. Setting thresholds on the basis of the monetary cost of training a model, rather than the total number of computational operations required—for example, the current FLOP threshold of 1e25 would be roughly equivalent to a training cost threshold of $7M (Sastry et al., 2024). Both types of thresholds ultimately measure the same thing: compute thresholds measure computational operations directly, while cost thresholds measure the monetary value associated with running the very same operations. While cost thresholds appear more intuitive as a regulatory tool, they have two drawbacks relative to compute thresholds:

- **More complex threshold adjustments**. In addition to algorithmic efficiency, cost thresholds may need to be adjusted for compute-price performance, as FLOP is getting cheaper every year, roughly at a rate of approximately 1.35x per year (Hobbhahn et al., 2023)[20].
- **Measurement complexity**. Measuring training cost is complex and lacks standardisation. It is unclear which cost factors—like cost of talent, and cost of experiments—should be included in overall training cost. Moreover, cost measurements have large uncertainty bounds (Hobbhahn et al., 2023).

Overall, training cost thresholds seem like a feasible method for classification of GPAI models as GPAI models with systemic risk, and worth considering, but may be more difficult to implement than compute thresholds, with little upside.

Benchmark thresholds. Setting thresholds based on models' scores on common AI capability benchmarks like MMLU, HELM, and SuperGLUE. Relative to compute thresholds, benchmark thresholds have the advantage that they are accounting for capabilities directly, without relying on an imperfect compute-capability relationship, but a number of challenges remain[21]:

- **Lack of accuracy and comprehensiveness**. Benchmark results are sensitive to inputs, such as prompt phrasing, context setting, and reasoning approaches, and lack generality, as most

[20] This may be less problematic if the goal of the threshold is tracking the current frontier, rather than a particular level of capabilities.

[21] See (Eriksson et al., 2025) for an interdisciplinary review of current issues with AI benchmarks.

benchmarks focus on narrow domains instead of real-world applications that require multiple capabilities. It is also difficult to measure the absence of capabilities, as models may fail particular tasks with inadequate elicitation, even if they possess the underlying capability.

- **Ease of circumvention**. Models can be designed or prompted to underperform on benchmarks by selective training on benchmark-like tasks, and other strategies. This is known as "sandbagging" (van der Weij et al., 2024). Moreover, there are concerns about verification, as an evaluated model may not match the deployed version.
- **Implementation challenges**. Capabilities can only be measured post-development, so evaluation of capabilities before training is not possible. Running benchmarks is also somewhat resource intensive, due to costs for computation, time, and expertise; and might provide a false sense of completeness to policymakers if results are not interpreted correctly.

Some of these concerns could be mitigated by using a diverse "basket of benchmarks" that measures different aspects of capabilities. The benchmarking approach outlined earlier in this report takes this route, and goes even further, by aggregating multiple benchmark scores into a single score using principal component analysis (Ruan et al., 2024). Other mitigation measures include standardised prompting protocols and clear evaluation timing and versioning requirements. Regulators could also consider using benchmarks in conjunction with training compute and other metrics.

7.4 Additional Synthetic Data Considerations

Contemporary training techniques, particularly synthetic training data generation, present significant challenges for compute attribution. While these techniques can enhance model capabilities, they bring the complexity of having three compute costs associated with them: (1) the training cost of the model that generated the synthetic data, (2) the inference compute to create the data in the first place and (3) the compute spent for the forward and backward pass during the primary model training. In contrast, human-generated training data misses the first two costs, making accounting for it much simpler. While our framework around cumulative compute definition effectively negates the question of whether to account for general-purpose synthetic data at all, we would like to highlight the complexities that arise if one nonetheless chooses to do so.

Consider a language model generating training examples: even accounting the compute spent per sample becomes problematic due to rejection sampling (Bai et al., 2022), where multiple samples may be generated and filtered to produce a single high-quality training example. This opens up the question of whether to account for the cost for all of the created samples regardless of the filtering or just the single sample generation cost. In practice, a model might generate 10 responses and select the best one, or generate 100 responses and select the top 1% based on quality filters. This variable rejection rate makes it impossible to establish a consistent compute-per-sample relationship, as the same synthetic data point might require vastly different amounts of compute depending on quality thresholds and sampling parameters.

Beyond measurement challenges, synthetic data is fundamentally temporally decoupled from the training process. The generated data retains its value independent of the training process, can be stored for future use, and often proves valuable for training entirely different models. This persistence of value suggests the compute used in generation should be considered a separate process from the training itself, more akin to dataset creation than model training. Projects like Ultrafeedback (Cui et al., 2023), which utilised 17 different models (many closed-source with undisclosed compute costs) to create preference training datasets, illustrate the difficulty of attributing generator model training or inference costs to the final model.

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List of abbreviations and definitions

AI Artificial Intelligence

CSP Cloud Service Provider

DPO Direct Preference Optimisation

FLOP Floating Point Operations

FMF Frontier Model Forum

GAN Generative Adversarial Networks

GPAI General-Purpose Artificial Intelligence

GPAI models General-Purpose Artificial Intelligence models

GPAISRs/GPAISR General-Purpose Artificial Intelligence models with Systemic Risk

IaaS Infrastructure as a Service

KYC Know Your Customer

MFU Model FLOP Utilisation

MCTS Monte Carlo Tree Search

OOM Order of magnitude

RL Reinforcement Learning

RLHF Reinforcement Learning from Human Feedback

SFT Supervised Fine-Tuning

SGD Standard Gradient Descent

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Annexes

Annex 1. Guidance on How to Measure

This annex outlines methodologies for measuring and verifying training compute for regulatory purposes. We build upon the guidance published by the Frontier Model Forum (FMF) (Frontier Model Forum, 2024), which represents a consensus among several leading AI developers, while extending this framework to address broader cumulative compute considerations relevant to the EU AI Act.

The FMF guidance provides a useful baseline that has industry support, but requires supplementation to fully address the concept of "cumulative" training compute and to account for evolving training methodologies.

A.1.1 Measurement Approaches

We recommend using two complementary approaches for estimating training compute, with the final determination based on the harmonic mean of these estimates to provide a more robust measurement:

1. Hardware-Based Approach

This approach measures compute directly based on hardware utilisation:

$$\textbf{FLOP} = \textbf{Number of Accelerators} \times \textbf{Peak FLOP/s} \times \textbf{Training Hours} \times \textbf{Utilisation Rate}$$

Where:

- **Number of Accelerators**: Count of GPUs/TPUs used during training.
- **Peak FLOP/s**: Theoretical maximum operations per second for the specific numerical format used (e.g., FP16, FP8).
- **Training Hours**: Total duration of the training process.
- **Utilisation Rate**: Percentage of theoretical peak performance achieved (typically 20-60% for large models).

2. Parameter-Based Approach

This approach estimates compute based on model architecture and training data:

$$\textbf{FLOP} = 6 \times \textbf{N} \times \textbf{D}$$

Where:

- **N**: Number of active model parameters.
- **D**: Number of training tokens processed.

For more complex architectures (e.g., mixture-of-experts, sparse models), N should be adjusted to reflect only active parameters used in each forward pass. Frozen parameters reduce the number operations from six to four as they skip both gradient computation and minibatch accumulation, yielding (2 x active-parameters + 4 x total-parameters) x D.

A.1.2 Definition of Cumulative Compute

Building on the FMF guidance, we define cumulative compute as the total computation directly contributing to parameter updates in the final deployed model. This includes two primary categories:

1. **Direct Parameter Updates:** Operations that update the final model's parameters through forward and backward propagation. These include matrix multiplications, gradient calculations, and optimisation steps that directly modify the model's weights and are unambiguously responsible for enhancing its capabilities.
2. **Model-Dependent Processes:** Computation that generates inputs for gradient updates that are specifically dependent on the current model state. These processes produce training signals (such as rewards, labels, or intermediate representations) that are specifically tailored to the model's evolving capabilities and cannot be easily repurposed for other models. Examples include reinforcement learning loops where the model's outputs determine subsequent training data or distillation processes where model-specific logits guide training.

From a model development timeline perspective, this includes:

1. **Pre-training compute:** The primary computation used during initial model development, typically involving large-scale self-supervised learning on diverse datasets.
2. **Post-training enhancements:** Compute used for fine-tuning, reinforcement learning from human feedback (RLHF), and other adjustments that directly modify model parameters to refine capabilities or align with human preferences.

Operations may be excluded from cumulative compute calculations when they:

1. Do not directly update model parameters in the final deployed model.
2. Create outputs never used as inputs to gradient computations affecting the deployed model.
3. Could be performed at any point without affecting training outcomes.

Based on these criteria, the following are typically excluded:

— Data preprocessing operations.

— Training of auxiliary models used for data generation.

— Inference computation for synthetic data generation.

— Discarded branches or failed experiments.

For ambiguous techniques, determine whether to include the computation by answering these questions:

Based on these criteria, the following are typically excluded:

— Data preprocessing operations.

— Training of auxiliary models used for data generation.

— Inference computation for synthetic data generation.

— Discarded branches or failed experiments.

For ambiguous techniques, determine whether to include the computation by answering these questions:

1. Does the computation directly update the final model's parameters?
2. Does the computation create model-specific inputs that depend on the current model state?

If the answer to ANY of questions 1 or 2 is "yes", the computation should be included in cumulative compute calculations. Otherwise, it should be excluded.

A.1.3 More Details

— **Numerical Precision:** All operations should be counted equally regardless of precision (e.g., FP32, FP16, FP8). While lower precision formats enable more operations per second on the same hardware, when comparing models trained with the same total FLOP count, precision differences produce only marginal capability differences.

— **Counted Operations:** Matrix multiplications are the primary operation in AI training. Element-wise and normalisation operations make up an insignificant amount of the total compute and can be omitted.

— **Synthetic Data:** We recommend that the computation used to generate reusable synthetic training data should be excluded from cumulative compute calculations (though see considerations under "Synthetic data" in Section 2.5). This includes both:

- Training compute for models that generate synthetic data.
- Inference compute used during the actual data generation process.

As part of this recommendation, only the computation used when the primary model processes this synthetic data during its own training would be counted.

— **Multi-modality:**

- Training compute for each component should be aggregated if the components are integrated during training.
- If separately trained models are combined post-training, the compute for training each component should be included in the cumulative total.

— **Sparse Models:** For sparse architectures like mixture-of-experts:

- Count only operations involving active parameters in each forward pass.
- For hardware-based estimation, adjust utilisation rates to reflect actual compute usage.

— **Model Merging:** Compute for merged models should be counted based on unique contributions. For checkpoint averaging within a single training run, only the incremental compute added between checkpoints should be counted, as this represents standard training with periodic averaging. In the case of models trained on different data distributions, known as "model soup", the compute associated with any shared base model should be counted once, while the unique compute for each specialised model should be cumulatively added after merging.

A.1.4 Required Documentation

For regulatory verification, providers should maintain and be prepared to provide:

1. Detailed compute capacity allocation (cluster specifications for each training phase).
2. Model efficiency metrics (utilisation rates measured and based on scaling experiments).
3. Compute estimation methodology with underlying assumptions.
4. Training duration.
5. Architecture specifications including parameter counts and activation patterns.

Our suggestion for the guidance on measuring training compute:

— The proposed guidance is our suggestion but subject to the other challenges and discussion points outlined in this memo. Clarifying them will inform this guidance.

Challenges for the guidance on measuring training compute:

A. **Complex Multi-Model Architectures:** How should training setups involving multiple interacting models be measured consistently? Current frameworks do not adequately address compute attribution across complex model pipelines.

B. **Reasoning-Focused Training:** We have limited insight into how reinforcement learning steps that enhance reasoning capabilities work. In DeepSeek's case, this phase represented a significant portion of their total compute expenditure. Additional technical details are needed to provide definitive guidance on these increasingly important training phases.

C. **Synthetic Data Equivalence:** Our measurement approach should not create artificial distinctions between companies generating synthetic data in-house versus those acquiring it externally. Both ultimately expend compute to train on this data—representing a form of algorithmic efficiency improvement similar to other data quality enhancements.

D. **Distillation Attribution:** Model distillation techniques (where knowledge transfers from "teacher" to "student" models) present accounting challenges. Unlike synthetic data generation, distillation typically involves direct access to internal model representations, logits, or other outputs. This creates measurement questions distinct from synthetic data generation that require specialised consideration.

E. **Measurement Standardisation:** The evolving nature of AI training techniques requires ongoing adaptation of measurement methodologies. How can we establish standards robust enough to accommodate emerging techniques while remaining consistent enough for regulatory purposes?

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